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ELECTRONIC DEVICE COMPRISING ANTENNA

Abstract

An electronic device is provided. The electronic device includes a wireless communication circuit, an antenna structure including a printed circuit board electrically connected to the wireless communication circuit, wherein the printed circuit board includes a first surface, a second surface facing an opposite direction from the first surface, and an antenna element disposed on the first surface or disposed inside the printed circuit board closer to the first surface than the second surface, a non-conductive dielectric facing the first surface and overlapping with the antenna element when viewed from above the first surface, and a conductive path at least a portion of which is disposed on the printed circuit board and which is electrically connected to the wireless communication circuit, wherein the conductive path is in physical contact with the non-conductive dielectric.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/015068, filed on Sep. 27, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0122501, filed on Sep. 27, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0141805, filed on Oct. 28, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to an electronic device comprising an antenna.

2. Description of Related Art

[0003] With the advancement of wireless communication technology, electronic devices are being used widely in our daily lives, and the use of content is increasing. Electronic devices contain multiple antennas to support various communication technologies.

[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0005] As the range of applications available expands, the number of antennas included in electronic devices is increasing. Electronic devices are becoming thinner while adding components for various functions, making it difficult to dispose antennas in limited spaces while reducing electrical influences with various elements within the electronic device and ensuring coverage (or communication range) and radiation performance for the desired frequency band.

[0006] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device including an antenna for securing or improving coverage and/or antenna radiation performance.

[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0008] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a wireless communication circuit, an antenna structure including a printed circuit board electrically connected to the wireless communication circuit, wherein the printed circuit board includes a first surface, a second surface facing an opposite direction from the first surface, and an antenna element disposed on the first surface or disposed inside the printed circuit board closer to the first surface than the second surface, a non-conductive dielectric facing the first surface and overlapping with the antenna element when viewed from above the first surface, and a conductive path at least a portion of which is disposed on the printed circuit board and which is electrically connected to the wireless communication circuit, wherein the conductive

path is in physical contact with the non-conductive dielectric.

[0009] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a wireless communication circuit, an antenna structure, a non-conductive dielectric, and a conductive path. The antenna structure includes a printed circuit board electrically connected to the wireless communication circuit. The printed circuit board includes a first surface and a second surface facing in the opposite direction from the first surface. The printed circuit board includes a first antenna element disposed on the first surface or disposed, closer to the first surface than the second surface, inside the printed circuit board. The printed circuit board includes a second antenna element disposed closer to the first surface than the first antenna element. The non-conductive dielectric may face the first surface and overlap the first antenna element and the second antenna element when viewed from above the first surface. The conductive path may be at least partially disposed on the printed circuit board and electrically connected to the wireless communication circuit. The conductive path may be in physical contact with the non-conductive dielectric.

[0010] An electronic device including an antenna according to an embodiment of the disclosure secures or improves coverage and/or antenna radiation performance by using a dielectric resonator.

[0011] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0013] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;

[0014] FIG. 2 is a block diagram illustrating an electronic device in a network environment including a plurality of cellular networks according to an embodiment of the disclosure;

[0015] FIG. 3 are diagrams illustrating a foldable electronic device in an unfolded state according to an embodiment of the disclosure;

[0016] FIG. 4 are diagrams illustrating a foldable electronic device in a folded state according to an embodiment of the disclosure;

[0017] FIGS. 5 and 6 are diagrams illustrating an antenna module according to various embodiments of the disclosure;

[0018] FIGS. 7 and 8 are diagrams illustrating beam patterns generated from an antenna module according to various embodiments of the disclosure;

[0019] FIG. 9 are diagrams illustrating the distribution of radiation fields generated from the first radiating portion according to an embodiment of the disclosure;

[0020] FIG. 10 are graphs illustrating radiation patterns of a first radiating portion according to an embodiment of the disclosure;

[0021] FIG. 11 are heat maps illustrating wave energy according to a feeding to an antenna module in free space according to an embodiment of the disclosure;

[0022] FIG. 12 is an exploded perspective view of an antenna module according to an embodiment of the disclosure;

[0023] FIG. 13 is a perspective view of an antenna module according to an embodiment of the disclosure;

[0024] FIG. 14 is a cross-sectional view of an antenna module as viewed along line A-A' in FIG. 13

according to an embodiment of the disclosure;

[0025] FIG. **15** is a cross-sectional view of an antenna module of another embodiment that is a variation of the embodiment of FIG. **14** according to an embodiment of the disclosure;

[0026] FIG. **16** are graphs illustrating radiation patterns of a first radiating portion according to the permittivity of the first dielectric in the embodiment of FIG. **13** according to an embodiment of the disclosure;

[0027] FIG. **17** are graphs illustrating radiation patterns of a first radiating portion along the length of the first conductive portion extending in a first direction toward the first surface in the embodiment of FIG. **14** according to an embodiment of the disclosure;

[0028] FIG. **18** is a diagram illustrating an antenna module according to an embodiment of the disclosure;

[0029] FIG. **19** is a cross-sectional view of an antenna module as viewed along line B-B' in FIG. **18** according to an embodiment of the disclosure;

[0030] FIG. **20** are heat maps illustrating wave energy for electromagnetic waves radiated by various feedings to the antenna module of FIG. **18** according to an embodiment of the disclosure;

[0031] FIG. **21** is a block diagram illustrating a communication system included in an electronic device according to an embodiment of the disclosure; and

[0032] FIG. **22** is a cross-sectional view illustrating a portion of an electronic device according to an embodiment of the disclosure.

[0033] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures

DETAILED DESCRIPTION

[0034] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0035] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0036] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0037] By the term “substantially” it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

[0038] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0039] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry

performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a Wi-Fi chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0040] FIG. 1 is a block diagram of an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0041] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an external electronic device **102** through a first network **198** (e.g., a short-range wireless communication network), or at least one of an external electronic device **104** or a server **108** through a second network **199** (e.g., a long-range wireless communication network). The electronic device **101** may communicate with the external electronic device **104** through the server **108**. The electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, and/or an antenna module **197**. In various embodiments of the disclosure, at least one (e.g., the connection terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments of the disclosure, some of the components may be implemented as single integrated circuitry. For example, the sensor module **176**, the camera module **180**, or the antenna module **197** may be implemented as embedded in single component (e.g., the display module **160**).

[0042] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. As at least part of the data processing or computation, the processor **120** may load a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. The processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. Additionally or alternatively, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0043] The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., a sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). The auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., a neural network processing device) may include a hardware structure specified for processing an artificial intelligence model. The artificial intelligence model may be created through machine learning. Such learning may be performed, for example, in the electronic

device **101** itself on which the artificial intelligence model is performed, or may be performed through a separate server (e.g., the server **108**). The learning algorithms may include, for example, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning, but is not limited thereto. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be any of a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent DNN (BRDNN), a deep Q-network, or a combination of two or more of the above-mentioned networks, but is not limited the above-mentioned examples. In addition to the hardware structure, the artificial intelligence model may additionally or alternatively include a software structure.

[0044] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** and/or the non-volatile memory **134**.

[0045] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, and/or an application **146**.

[0046] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0047] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for incoming calls. The receiver may be implemented as separate from, or as part of the speaker.

[0048] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. The display module **160** may include touch circuitry (e.g., a touch sensor) adapted to detect a touch, or sensor circuitry (e.g., a pressure sensor) adapted to measure the intensity of force incurred by the touch.

[0049] The audio module **170** may convert a sound into an electrical signal and vice versa. The audio module **170** may obtain the sound through the input module **150**, or output the sound through the sound output module **155** or a headphone of an external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0050] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. The sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0051] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly. The interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, and/or an audio interface. The connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the external electronic device **102**). The connecting terminal **178** may

include, for example, an HDMI connector, a USB connector, an SD card connector, and/or an audio connector (e.g., a headphone connector).

[0052] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. The haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0053] The camera module **180** may capture a still image or moving images. The camera module **180** may include one or more lenses, image sensors, ISPs, or flashes.

[0054] The power management module **188** may manage power supplied to or consumed by the electronic device **101**. The power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0055] The battery **189** may supply power to at least one component of the electronic device **101**. The battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, and/or a fuel cell.

[0056] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication through the established communication channel. The communication module **190** may include one or more CPs that are operable independently from the processor **120** (e.g., the AP) and supports a direct (e.g., wired) communication or a wireless communication. The communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device through the first network **198** (e.g., a short-range communication network, such as BLUETOOTH, wireless-fidelity (Wi-Fi) direct, or IR data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5th generation (5G) network, a next generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.

[0057] The wireless communication module **192** may support a 5G network, after a 4th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support high-speed transmission of high-capacity data (i.e., enhanced mobile broadband (eMBB)), minimization of terminal power and connection of multiple terminals (massive machine type communications (mMTC)), or high reliability and low latency (ultra-reliable and low-latency communications (URLLC)). The wireless communication module **192** may support a high-frequency band (e.g., a millimeter wave (mm Wave) band) to achieve, for example, a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance in a high-frequency band, such as beamforming, massive multiple-input and multiple-output (MIMO), full-dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large-scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the external electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment of the disclosure, the wireless communication module

192 may support a peak data rate for implementing eMBB (e.g., 20 Gbps or more), loss coverage for implementing mMTC (e.g., 164 dB or less), or U-plane latency for realizing URLLC (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL) or 1 ms or less for round trip).

[0058] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. The antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). The antenna module **197** may include a plurality of antennas (e.g., an antenna array). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device through the selected at least one antenna. Another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0059] According to various embodiments of the disclosure, the antenna module **197** may form a mmWave antenna module. According to an embodiment of the disclosure, the mm Wave antenna module may include a PCB, an RFIC that is disposed on or adjacent to a first surface (e.g., the bottom surface) of the PCB and is capable of supporting a predetermined high-frequency band (e.g., a mm Wave band), and a plurality of antennas (e.g., array antennas) that is disposed on or adjacent to a second surface (e.g., the top surface or the side surface) of the PCB and is capable of transmitting or receiving a signal of the predetermined high-frequency band.

[0060] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0061] Commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** through the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. All or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102** or **104**, or the server **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide an ultra-low delay service using, for example, distributed computing or MEC. In another embodiment of the disclosure, the external electronic device **104** may include an internet of things (IoT) device. The server **108** may be an intelligent server using machine learning and/or neural networks. According to an embodiment of the disclosure, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to an intelligent service (e.g., smart home, smart city, smart car, or healthcare) on the basis of 5G communication technology or IoT-related technology.

[0062] An electronic device according to an embodiment of the disclosure may be one of various types of electronic devices. The electronic devices may include a portable communication device

(e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. However, the electronic device is not limited to any of those described above.

[0063] Various embodiments of the disclosure and the terms used herein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). If an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively,” as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or through a third element.

[0064] The term “module” may include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment of the disclosure, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0065] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0066] A method according to an embodiment of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online through an application store (e.g., PLAYSTORE™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0067] Each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. One or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. Operations performed by the module, the program,

or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0068] FIG. 2 is a block diagram **200** illustrating an electronic device **101** in a network environment including a plurality of cellular networks according to an embodiment of the disclosure.

[0069] Referring to FIG. 2, the electronic device **101** may include a first communication processor **212**, a second communication processor **214**, a first radio frequency integrated circuit (RFIC) **222**, a second RFIC **224**, a third RFIC **226**, a fourth RFIC **228**, a first radio frequency front end (RFFE) **232**, a second RFFE **234**, a first antenna module **242**, a second antenna module **244**, and/or an antenna **248**. The electronic device **101** may include a processor **120** and a memory **130**. The second network **199** may include a first cellular network **292** and a second cellular network **294**. In another embodiment, the electronic device **101** may further include at least one of the components described in FIG. 1, and the second network **199** may further include at least one other network. According to an embodiment, the first communication processor **212**, the second communication processor **214**, the first RFIC **222**, the second RFIC **224**, the fourth RFIC **228**, the first RFFE **232**, and the second RFFE **234** may form at least a portion of the wireless communication module **192**. In another embodiment, the fourth RFIC **228** may be omitted or may be included as a part of the third RFIC **226**.

[0070] The first communication processor **212** may support establishment of a communication channel of a band to be used for wireless communication with the first cellular network **292**, and legacy network communication through the established communication channel. According to various embodiments, the first cellular network may be a legacy network including a second generation (2G), a third generation (3G), a fourth generation (4G), or a long term evolution (LTE) network. The second communication processor **214** may support establishment of a communication channel corresponding to a designated band (e.g., about 6 GHz to about 60 GHz) among the bands to be used for wireless communication with the second cellular network **294**, and 5G network communication through the established communication channel. According to various embodiments, the second cellular network **294** may be a fifth generation (5G) network defined by 3GPP. Additionally, according to an embodiment, the first communication processor **212** or the second communication processor **214** may support establishment of a communication channel corresponding to another designated band (e.g., about 6 GHz or less) among the bands to be used for wireless communication with the second cellular network **294**, and 5G network communication through the established communication channel. According to an embodiment, the first communication processor **212** and the second communication processor **214** may be implemented in a single chip or a single package. According to various embodiments, the first communication processor **212** or the second communication processor **214** may be formed in a single chip or a single package with the processor **120**, the auxiliary processor **123**, or the communication module **190**.

[0071] The first RFIC **222** may, during transmission, convert a baseband signal generated by the first communication processor **212** into a radio frequency (RF) signal of about 700 MHz to about 3 GHz used in the first cellular network **292** (e.g., a legacy network). During reception, the RF signal may be acquired from the first cellular network **292** (e.g., a legacy network) through an antenna (e.g., the first antenna module **242**) and preprocessed through an RFFE (e.g., the first RFFE **232**). The first RFIC **222** may convert the preprocessed RF signal into a baseband signal so that it may be processed by the first communication processor **212**.

[0072] The second RFIC **224** may, during transmission, convert a baseband signal generated by the first communication processor **212** or the second communication processor **214** into an RF signal (hereinafter, a 5G Sub6 RF signal) of a Sub6 band (e.g., about 6 GHz or less) used in the second cellular network **294** (e.g., a 5G network). During reception, the 5G Sub6 RF signal may be

acquired from the second cellular network **294** (e.g., the 5G network) through an antenna (e.g., the second antenna module **244**) and preprocessed through an RFFE (e.g., the second RFFE **234**). The second RFIC **224** may convert the preprocessed 5G Sub6 RF signal into a baseband signal so that the preprocessed 5G Sub6 RF signal may be processed by a corresponding communication processor among the first communication processor **212** or the second communication processor **214**.

[0073] The third RFIC **226** may convert a baseband signal generated by the second communication processor **214** into an RF signal (hereinafter, “5G Above6 RF signal”) of a 5G Above6 band (e.g., about 6 GHz to about 60 GHz) to be used in the second cellular network **294** (e.g., a 5G network). During reception, the 5G Above6 RF signal may be acquired from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., the antenna **248**) and preprocessed through the third RFFE **236**. The third RFIC **226** may convert the preprocessed 5G Above6 RF signal into a baseband signal so that it can be processed by the second communication processor **214**. According to an embodiment, the third RFFE **236** may be formed as a part of the third RFIC **226**.

[0074] The electronic device **101** may, according to an embodiment, include a fourth RFIC **228** separately from or at least as a part of the third RFIC **226**. In such a case, the fourth RFIC **228** may convert a baseband signal generated by the second communication processor **214** into an RF signal (hereinafter, referred to as an IF signal) of an intermediate frequency band (e.g., about 9 GHz to about 11 GHz) and then transmit the IF signal to the third RFIC **226**. The third RFIC **226** may convert the IF signal into a 5G Above6 RF signal. During reception, the 5G Above6 RF signal may be received from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., antenna **248**) and converted into an IF signal by the third RFIC **226**. The fourth RFIC **228** may convert the IF signal into a baseband signal for processing by the second communications processor **214**.

[0075] According to an embodiment, the first RFIC **222** and the second RFIC **224** may be implemented as at least a portion of a single chip or a single package. According to an embodiment, the first RFFE **232** and the second RFFE **234** may be implemented as at least a portion of a single chip or a single package. According to an embodiment, at least one antenna module of the first antenna module **242** or the second antenna module **244** may be omitted or combined with another antenna module to process RF signals of corresponding multiple bands.

[0076] According to an embodiment, the third RFIC **226** and the antenna **248** may be disposed on the same substrate to form a third antenna module **246**. For example, the wireless communication module **192** or the processor **120** may be disposed on a substrate (e.g. a main PCB). In this case, a third RFIC **226** may be disposed on a portion (e.g., a lower surface) of a second substrate (e.g., a sub PCB) separate from the first substrate, and an antenna **248** may be disposed on another portion (e.g., a top surface) to form a third antenna module **246**. By disposing the third RFIC **226** and the antenna **248** on the same substrate, it is possible to reduce the length of a transmission line therebetween. This may reduce, for example, the loss (e.g., attenuation) of a signal in a high-frequency band (e.g., about 6 GHz to about 60 GHz) used for 5G network communication by the transmission line. As a result, the electronic device **101** may improve the quality or speed of communication with the second cellular network **294** (e.g., a 5G network).

[0077] According to an embodiment, the antenna **248** may be formed as an antenna array comprising a plurality of antenna elements that may be used for beamforming. In this case, the third RFIC **226** may include, for example, a plurality of phase shifters **238** corresponding to the plurality of antenna elements as part of the third RFFE **236**. During transmission, each of the plurality of phase shifters **238** may shift the phase of a 5G Above6 RF signal to be transmitted to an external source (e.g., a base station of a 5G network) of the electronic device **101** through its corresponding antenna element. During reception, each of the plurality of phase shifters **238** may shift the phase of a 5G Above6 RF signal received from the external source through its corresponding antenna element to the same or substantially the same phase. This enables

transmission or reception through beamforming between the electronic device **101** and the external source.

[0078] The second cellular network **294** (e.g., a 5G network) may operate independently (e.g., Stand-Alone (SA)) or connected (e.g., Non-Stand Alone (NSA)) to the first cellular network **292** (e.g., a legacy network). For example, the 5G network may only have an access network (e.g., a 5G radio access network (RAN) or next generation RAN (NG RAN)) and no core network (e.g., next generation core (NGC)). In such a case, the electronic device **101** may access an external network (e.g., the Internet) under the control of the core network (e.g., evolved packet core (EPC)) of the legacy network after accessing the access network of the 5G network. Protocol information for communicating with a legacy network (e.g., LTE protocol information) or protocol information for communicating with a 5G network (e.g., New Radio (NR) protocol information) may be stored in the memory **230** and accessed by other components (e.g., the processor **120**, the first communication processor **212**, or the second communication processor **214**).

[0079] FIG. **3** are diagrams illustrating a foldable electronic device **3** in an unfolded state (or unfolding state, or flat state) according to an embodiment of the disclosure. FIG. **4** are diagrams illustrating a foldable electronic device **3** in a folded state (or folding state) according to an embodiment of the disclosure.

[0080] Referring to FIGS. **3** and **4**, the foldable electronic device **3** may include a foldable housing **30**, a first display module (e.g., a flexible display module or a foldable display module) **34**, and/or a second display module **35**. In various embodiments, the foldable electronic device **3** may be the electronic device **101** of FIG. **1**.

[0081] According to an embodiment, the foldable housing **30** may include a first housing (or a first housing portion or a first housing structure) **31**, a second housing (or a second housing portion or a second housing structure) **32**, a hinge housing **33**, and/or a hinge portion (not shown separately). The first housing **31** and the second housing **32** may be connected through the hinge portion and may be rotatable corresponding to each other with respect to the hinge portion. The hinge portion may include one or more hinges or hinge modules (or hinge assemblies).

[0082] According to an embodiment, the display area **34A** of the first display module **34** may include a first display area (or first active area or first screen area) {circle around (1)}, a second display area (or second active area or second screen area) {circle around (2)}, and a third display area (or third active area or third screen area) {circle around (3)} as active areas of the first display module **34** capable of displaying an image. The first display area {circle around (1)} may be positioned corresponding to the first housing **31**. The second display area {circle around (2)} may be positioned corresponding to the second housing **32**. The third display area {circle around (3)} may be a portion connecting the first display area {circle around (1)} and the second display area {circle around (2)} of the first display module **34**. The third display area {circle around (3)} may be positioned corresponding to a hinge portion. The first display area {circle around (1)} may be disposed in the first housing **31**, and the shape of the first display area {circle around (1)} may be maintained by the support of the first housing **31**. The second display area {circle around (2)} may be disposed in the second housing **32**, and the shape of the second display area {circle around (2)} may be maintained by the support of the second housing **32**. The first display area {circle around (1)} and the second display area {circle around (2)} may be provided substantially flat, for example. The unfolded state of the foldable electronic device **3** may be a state in which the third display area {circle around (3)} is disposed substantially flat.

[0083] According to an embodiment, in the unfolded state of the foldable electronic device **3**, the first display area {circle around (1)} and the second display area {circle around (2)} may form an angle of about 180 degrees, and the display area **34A** including the first display area {circle around (1)}, the second display area {circle around (2)}, and the third display area {circle around (3)} may be provided (or disposed) in a substantially flat form. Because of the relative positions between the first display area {circle around (1)} disposed in the first housing **31** and the second display area

{circle around (2)} disposed in the second housing 32 in the unfolded state of the foldable electronic device 3, the third display area {circle around (3)} connecting the first display area {circle around (1)} and the second display area {circle around (2)} may be disposed substantially flat. For example, in the unfolded state of the foldable electronic device 3, the third display area {circle around (3)} may be pulled from both sides by the first display area {circle around (1)} and the second display area {circle around (2)}, and the pulling force may be provided so as to reduce damage to the third display area {circle around (3)} while flatly disposing the third display area {circle around (3)}. For example, the third display area {circle around (3)} may be provided with an extended width so as to be flatly disposed while reducing stress caused by being pulled by the first display area {circle around (1)} and the second display area {circle around (2)} in the unfolded state of the foldable electronic device 3.

[0084] The coordinate axes are illustrated on the basis of the first housing 31, and for example, the +z-axis direction may be interpreted as the direction in which the flat first display area {circle around (1)} of the foldable electronic device 3 faces, the y-axis direction may be interpreted as the direction parallel to the center line A of the foldable electronic device 3, and the x-axis direction may be interpreted as the direction perpendicular to the center line A of the foldable electronic device 3.

[0085] According to an embodiment, in the unfolded state of the foldable electronic device 3, the hinge portion may support the third display area {circle around (3)}. In the unfolded state of the foldable electronic device 3, when an external force (e.g., external pressure such as a touch input using a user's finger or a touch input using an electronic pen) is applied to the third display area {circle around (3)}, the hinge portion may contribute to reducing the sagging phenomenon of the third display area {circle around (3)} so that the third display area {circle around (3)} can be maintained flat. When an external impact is applied to the foldable electronic device 3 in the unfolded state because of a reason such as a drop, the hinge portion may be configured to reduce the impact of the external impact on the third display area {circle around (3)}. The hinge portion may support the third display area {circle around (3)} such that the third display area {circle around (3)} can be disposed flatly without sagging in the unfolded state of the foldable electronic device 3, thereby reducing the phenomenon of creasing.

[0086] According to an embodiment, the foldable electronic device 3 may be provided in an infolding manner in which the display area 34A of the first display module 34 is folded inward. A surface of the foldable electronic device 3 on which the display area 34A of the first display module 34 is visible may be interpreted as the 'front surface of the foldable electronic device 3', and a surface of the foldable electronic device 3 facing in the opposite direction from the front surface of the foldable electronic device 3 may be interpreted as the 'rear surface of the foldable electronic device 3'. The foldable electronic device 3 may be configured such that the front surface or the first display module 34 may be folded inward.

[0087] According to an embodiment, FIG. 4 illustrates a fully folded state of the foldable electronic device 3 in which the first housing 31 and the second housing 32 are disposed so that they no longer come close to each other. In the fully folded state of the foldable electronic device 3, the first display area {circle around (1)} and the second display area {circle around (2)} may be positioned to face each other, and the third display area {circle around (3)} may be disposed in a bended form. In the fully folded state of the foldable electronic device 3, the angle between the first housing 31 and the second housing 32 (or the angle between the first display area {circle around (1)} and the second display area {circle around (2)}) may be about 0 degree to about 10 degrees, and the display area 34A may be substantially invisible. Although not illustrated, the intermediate state of the foldable electronic device 3 may be a state between the unfolded state and the fully folded state, or a less folded state compared to the fully folded state. For example, in the case of an intermediate state in which the angle between the first housing 31 and the second housing 32 is greater than a certain angle, a usage environment in which the user does not have substantial difficulty in using

the display area **34A** may be provided. Hereinafter, the ‘folded state of the foldable electronic device **3**’ described in the disclosure may refer to a fully folded state as opposed to an intermediate state of a less folded state.

[0088] According to an embodiment, when viewing the unfolded state of the foldable electronic device **3**, the display area **34A** of the first display module **34** may be provided in a symmetrical form with respect to the center line A of the foldable electronic device **3**. The center line A may correspond to the middle of the width of the third display area {circle around (3)} extended from the first boundary between the first display area {circle around (1)} and the third display area {circle around (3)} to the second boundary between the second display area {circle around (2)} and the third display area {circle around (3)} when viewing the unfolded state of the foldable electronic device **3**. The third display area {circle around (3)} disposed in a bended form in the folded state of the foldable electronic device **3** may be in a substantially symmetrical form with respect to the center line A of the foldable electronic device **3**.

[0089] According to an embodiment, when viewing the unfolded state of the foldable electronic device **3**, the display area **34A** of the first display module **34** may be substantially rectangular. The display area **34A** may include a first edge B1, a second edge B2, a third edge B3, and a fourth edge B4. The first edge B1 and the second edge B2 may be substantially parallel to the center line A. The third edge B3 may connect one end of the first edge B1 and one end of the second edge B2, and the fourth edge B4 may connect the other end of the first edge B1 and the other end of the second edge B2. The first display area {circle around (1)} may include the first edge B1, a portion of the third edge B3, and a portion of the fourth edge B4. The second display area {circle around (2)} may include a second edge B2, a portion of the third edge B3, and a portion of the fourth edge B4. The third display area {circle around (3)} may include a portion of the third edge B3 and a portion of the fourth edge B4. In the folded state of the foldable electronic device **3**, the first edge B1 and the second edge B2 may be substantially overlapped and aligned. In the folded state of the foldable electronic device **3**, a portion of the third edge B3 included in the first display area {circle around (1)} and a portion of the third edge B3 included in the second display area {circle around (2)} may be substantially overlapped and aligned. In the folded state of the foldable electronic device **3**, a portion of the fourth edge B4 included in the first display area {circle around (1)} and a portion of the fourth edge B4 included in the second display area {circle around (2)} may be substantially overlapped and aligned.

[0090] According to an embodiment, the first display module **34** may include a first border area (not shown separately) extended from the first display area {circle around (1)}. The combination of the first display area {circle around (1)} and the first border area of the first display module **34** may be defined or interpreted as a ‘first area’, and the first area may be disposed in the first housing **31**. The first border area may be covered by the first housing **31** and may not be exposed to the outside. The first display module **34** may include a second border area (not shown separately) extended from the second display area {circle around (2)}. The combination of the second display area {circle around (2)} and the second border area of the first display module **34** may be defined or interpreted as a ‘second area’, and the second area may be disposed in the second housing **32**. The second border area may be covered by the first housing **31** and not be exposed to the outside.

[0091] According to an embodiment, the first display module **34** may include a third border area (not shown separately) extending from the third display area {circle around (3)}. A combination of the third display area {circle around (3)} and the third border area among the first display modules **34** may be defined or interpreted as a ‘third area’, and the third area may be positioned corresponding to the hinge portion. The third border area may be covered by a material (or member) disposed in the third border area and may not be exposed to the outside. The material disposed in the third border area may be exposed to the outside in an unfolded state of the foldable electronic device **3** to provide a portion of the appearance of the foldable electronic device **3**. The material disposed in the third border area may include a flexible material that may reduce a

decrease in the bendability of the third display area {circle around (3)} (or the third area) when it is disposed in a bended form in a folded state of the foldable electronic device 3.

[0092] According to an embodiment, the first housing 31 may include a first frame (or a first frame structure or a first framework) 311, and/or a first cover 312 disposed on the first frame 311.

[0093] According to an embodiment, the first frame 311 may include a first support member 3111 and a first side (or a first side surface, a first lateral member, a first side surface structure, or a first side surface bezel structure) 3112. The support member 3111 and the first side 3112 may be connected. The first frame 311 may be provided in an integral form, for example, including the first support member 3111 and the first side 3112. The first support member 3111 may be referred to by various other terms such as a 'first bracket', a 'first support body', a 'first support member', or a 'first support structure' as an internal structure positioned inside the foldable electronic device 3 corresponding to the first housing 31. For example, the first display area {circle around (1)} may be disposed on the first support member 3111, and the first support member 3111 may support the first display area {circle around (1)}. The first display area {circle around (1)} and the first cover 312 may be positioned on opposite sides with the first support member 3111 interposed therebetween. The first side 3112 may be disposed to at least partially surround a space between the first display area {circle around (1)} and the first cover 312. The first side 3112 may be disposed along an edge of the first display area {circle around (1)} of the first display module 34. The first side 3112 may provide a first side surface of the foldable electronic device 3 corresponding to the first display area {circle around (1)} of the foldable electronic device 3. The first display area {circle around (1)} may provide one surface (e.g., the first front surface or the first front surface area) of an outer surface of the foldable electronic device 3, and the first cover 312 may provide the other surface (e.g., the first rear surface or the first rear surface area) of the outer surface of the foldable electronic device 3 facing in a direction substantially opposite to the first display area {circle around (1)}. The first cover 312 may be referred to as a 'first rear cover' as an element that provides at least a portion of the first rear surface corresponding to the first housing 31 of the rear surface of the foldable electronic device 3. Various electrical components (or electronic parts), such as printed circuit boards or batteries, may be disposed on the first support 3111 between the first support 3111 and the first cover 312.

[0094] According to an embodiment, the second housing 32 may include a second frame (or a second frame structure or a second framework) 321, and/or a second cover 322 disposed on the second frame 321.

[0095] According to an embodiment, the second frame 321 may include a second support member 3211 and a second side (or a second side surface, a second lateral member, a second side surface structure, or a second side surface bezel structure) 3212. The support member 3211 and the second side 3212 may be connected. The second frame 321 may be provided in an integral form, for example, including the second support member 3211 and the second side 3212. The second support member 3211 may be referred to by various other terms such as a 'second bracket', a 'second support body', a 'second support member', or a 'second support structure' as an internal structure positioned inside the foldable electronic device 3 corresponding to the second housing 32. For example, the second display area {circle around (2)} may be disposed on the second support member 3211, and the second support member 3211 may support the second display area {circle around (2)}. The second display area {circle around (2)} and the second cover 322 may be positioned on opposite sides with the second support member 3211 therebetween. The second side 3212 may be disposed to at least partially surround a space between the second display area 2 and the second cover 322. The second side 3212 may be disposed along the edge of the second display area {circle around (2)} of the first display module 34. The second side 3212 may provide a second side surface of the foldable electronic device 3 corresponding to the second display area {circle around (2)} of the foldable electronic device 3. The second display area {circle around (2)} may provide one surface (e.g., the second front surface or the second front surface area) of the outer

surface of the foldable electronic device **3**, and the second cover **322** may provide the other surface (e.g., the second rear surface or the second rear surface area) of the outer surface of the foldable electronic device **3** facing in a direction substantially opposite to the second display area {circle around (2)}. The second cover **322** may be referred to as a ‘second rear cover’ as an element that provides at least a portion of the second rear surface corresponding to the second housing **32** of the rear surface of the foldable electronic device **3**. Various electrical components (or electronic parts), such as printed circuit boards or batteries may be disposed on the second support **3211** between the second support **3211** and the second cover **322**.

[0096] According to an embodiment, the foldable electronic device **3** may include one or more flexible printed circuit boards (not shown separately) disposed across the hinge portion. The one or more flexible printed circuit boards may electrically connect a first electrical element (e.g., a first printed circuit board (not shown separately)) disposed on a first frame **311** and a second electrical element (e.g., a second printed circuit board (not shown separately)) disposed on a second frame **321**.

[0097] According to various embodiments, the foldable electronic device **3** may further include a first inner support (not shown separately) disposed or coupled to the first frame **311**. For example, the first inner support may be positioned between the first cover **312** and the first support portion **3111** of the first frame **311**. The first inner support may be disposed on or coupled to the first support portion **3111** through various methods, such as screw fastening. The first inner support may cover and protect a component, such as a first printed circuit board (not shown separately) or a first battery (not shown separately), disposed between the first support portion **3111** and the first cover **312**. In various embodiments, the first frame **311** may be referred to as a ‘first front case’ and the first internal support may be referred to as a ‘first rear case’.

[0098] According to various embodiments, the first inner support may include a non-metallic material, and a conductive pattern utilized as an antenna radiator may be disposed or included in the first inner support.

[0099] According to various embodiments, the foldable electronic device **3** may further include a second inner support (not shown separately) disposed or coupled to the second frame **321**. For example, the second inner support may be positioned between the second cover **322** and the second support **3211** of the second frame **321**. The second inner support may be disposed on or coupled to the second support **3211** in various ways, such as screw fastening. The second inner support may cover and protect a component, such as a second printed circuit board (not shown separately) or a second battery (not shown separately), disposed between the second support **3211** and the second cover **322**. In various embodiments, the second frame **321** may be referred to as a ‘second front case’ and the second inner support may be referred to as a ‘second rear case’.

[0100] According to various embodiments, the second inner support may comprise a non-metallic material, and a conductive pattern utilized as an antenna radiator may be disposed or included in the second inner support.

[0101] According to an embodiment, the hinge housing (or hinge cover) **33** may be combined with a hinge module (not shown separately). When the foldable electronic device **3** is transitioned from an unfolded state to a folded state, because of a change in the relative position between the first frame **311** and the second frame **321** which are connected to each other through the hinge portion and a change in the state of the hinge portion (or hinge module) combined with the hinge housing **33**, a gap between the first housing **31** and the second housing **32** on the opposite side of the third display area {circle around (3)} of the first display module **34** may be opened, and the hinge housing **33** may be exposed to the outside through the opened gap. In the folded state of the foldable electronic device **3**, the hinge housing **33** may become a portion of the exterior that covers the interior of the foldable electronic device **3**. The hinge housing **33** may be more exposed to the outside in the completely folded state (see FIG. 4) than in the intermediate state of the foldable electronic device **3**. When the foldable electronic device **3** is transitioned from the folded state to

the unfolded state, because of a change in the relative position between the first frame **311** and the second frame **321** connected to each other through the hinge portion and a change in the state of the hinge portion (or hinge module) combined with the hinge housing **33**, the gap between the first housing **31** and the second housing **32** on the opposite side of the third display area {circle around (3)} of the first display module **34** may be closed, and the hinge housing **33** may be positioned in the inner space resulting from the combination of the first frame **311** and the second frame **321** and may not be exposed to the outside.

[0102] According to an embodiment, the second display module **35** may be disposed in a different direction from the first display module **34**. The second display module **35** may, for example, provide a display area (or screen area) facing in the opposite direction from the second display area {circle around (2)} of the first display module **34**.

[0103] According to an embodiment, the second cover **322** may include a light transmitting area (or opening) provided corresponding to the second display module **35**, and the second display module **35** may be visually visible through the light transmitting area of the second cover **322**. The second display module **35** may be provided in a form in which a transparent cover (a transparent plate that covers and protects the display, for example, a window) is omitted, and may be combined with the second cover **322** through an optically transparent adhesive material (for example, an optical clear adhesive (OCA), an optical clear resin (OCR), or a super view resin (SVR)).

[0104] According to various embodiments, the second display module **35** may be provided in a form including a second cover **322**. In this case, the second cover **322** may be interpreted as being excluded from the foldable housing **30**.

[0105] According to an embodiment, the foldable electronic device **3** may be configured to display an image through a second display module **35** instead of the first display module **34** in a folded state.

[0106] According to an embodiment, the first cover **312** may include a first curved area **312a** that is seamlessly extended and curved toward the first display area {circle around (1)} corresponding to the first edge B1 of the display area **34A**. The second cover **322** may include a second curved area **322a** that is seamlessly extended and curved toward the second display area {circle around (2)} corresponding to the second edge B2 of the display area **24A**. In the unfolded state or the folded state of the foldable electronic device **3**, the first curved area **312a** and the second curved area **322a** may be provided symmetrically to each other on opposite sides, thereby contributing to an attractive appearance.

[0107] According to an embodiment, the second display module **35** may include a flexible display that can be bent and disposed along the second curved area **322a**.

[0108] According to various embodiments, the first cover **312** may be provided substantially flat without the first curved area **312a**. The second cover **322** may be provided substantially flat without the second curved area **322a**. In this case, the second display module **35** may include a rigid display.

[0109] According to an embodiment, the foldable electronic device **3** may include one or more audio input modules (e.g., the input module **150** of FIG. 1), one or more audio output modules (e.g., the sound output module **155** of FIG. 1), one or more sensor modules (e.g., the sensor module **176** of FIG. 1), one or more camera modules (e.g., the first camera module **405**, the second camera module **406**, the third camera module **407**, and/or the fourth camera module **408**), one or more light-emitting modules (e.g., the light-emitting module **409**), one or more connection terminals, one or more key input modules (e.g., the input module **150** of FIG. 1), and/or an antenna module **5**. The foldable electronic device **3** may omit at least one of the components or additionally include other components. The positions or numbers of the components included in the foldable electronic device **3** are not limited to the illustrated examples and may vary.

[0110] Any one of the one or more sound input modules may, for example, include a microphone (or microphone module) (not shown separately) positioned inside the foldable electronic device **3**

corresponding to a microphone hole **401** provided on an exterior (e.g., a first side surface) of the foldable electronic device **3**. The positions or numbers of microphones and microphone holes corresponding to the microphones are not limited to the illustrated examples and may vary. In various embodiments, the foldable electronic device **3** may include a plurality of microphones (not shown separately) used to detect a direction of sound.

[0111] One of the one or more audio output modules may, for example, include a first speaker (or a first speaker module) (not shown separately) for multimedia playback (or for recording playback) positioned inside the foldable electronic device **3** corresponding to a first speaker hole **402** provided on an exterior surface (e.g., a second side surface) of the foldable electronic device **3**. Another one of the one or more audio output modules may, for example, include a second speaker (or a second speaker module) (e.g., a call receiver) (not shown separately) for calling positioned inside the foldable electronic device **3** corresponding to a second speaker hole (e.g., a receiver hole) **403** provided on an exterior surface (e.g., a first rear surface) of the foldable electronic device **3**. The positions or numbers of the speakers and the speaker holes corresponding to the speakers are not limited to the illustrated examples and may vary. In various embodiments, the microphone hole and the speaker hole may be implemented as a single hole (not shown separately). In various embodiments, a piezo speaker in which the speaker hole is omitted may be provided (not shown separately).

[0112] Any one of the one or more sensor modules may include an optical sensor (or optical sensor module) **404** positioned in the inner space of the second housing **32** corresponding to the second cover **322**. The optical sensor **404** may be positioned in alignment with an opening provided in the second display module **35** or may be at least partially inserted into the opening. External light may reach the optical sensor **404** through the second cover **322** and the opening provided in the second display module **35**. The optical sensor **404** may include, for example, a proximity sensor or an illuminance sensor. The number or location of the optical sensors is not limited to the illustrated examples and may vary.

[0113] According to various embodiments, the optical sensor **404** may be positioned in the inner space of the second housing **32** so as to overlap with at least a portion of the display area of the second display module **35** when viewed from above the second cover **322**. In this case, the optical sensor **404** or the location of the optical sensor **404** may be visually distinguished (or exposed) or invisible. The optical sensor **404** may be positioned, for example, on the rear surface of the second display module **35** or below or beneath the second display module **35**. In various embodiments, the optical sensor **404** may be positioned aligned with or at least partially inserted into a recess provided in the rear surface of the second display module **35**. A portion of the second display module **35** that at least partially overlaps with the optical sensor **404** may include a different pixel structure and/or wiring structure from other portions (not shown separately). For example, a portion of the second display module **35** that at least partially overlaps with the optical sensor **404** may have a different pixel density (e.g., number of pixels per unit area) from other portions. A pixel structure and/or wiring structure formed in a portion of the second display module **35** that at least partially overlaps with the optical sensor **404** may reduce light loss between the outside and the optical sensor **404**. As another example, a plurality of pixels may not be disposed in a portion of the second display module **35** that at least partially overlaps with the optical sensor **404**.

[0114] According to various embodiments, various other sensors (not shown separately), not limited to an optical sensor **404**, such as a proximity sensor or an illuminance sensor, may be positioned corresponding to an opening provided in the second display module **35**, or positioned on the rear surface of the second display module **35** or below or beneath the second display module **35**. For example, an optical, electrostatic, or ultrasonic biometric sensor (e.g., a fingerprint sensor) may be positioned corresponding to an opening provided in the second display module **35**, or positioned on the rear surface of the second display module **35** or below or beneath the second display module **35**.

[0115] According to various embodiments, various sensors (or sensor modules) may be positioned corresponding to openings provided in the first display module **34**, or positioned on the rear surface of the first display module **34**, or below first display module **34** (not shown separately).

[0116] According to an embodiment, the first camera module **405** may be positioned in the inner space of the second housing **32** corresponding to the second cover **322**. For example, the first camera module **405** may be positioned in alignment with an opening provided in the second display module **35** or may be at least partially inserted into the opening. External light may reach the first camera module **405** through the opening of the second cover **322** and the second display module **35**. The opening of the second display module **35** aligned with or overlapping with the first camera module **405** may be provided in the form of a through hole, as in the illustrated example. In various embodiments, the opening of the second display module **35** aligned with or overlapping with the first camera module **405** may be provided as a notch (not illustrated separately).

[0117] According to various embodiments, the first camera module **405** may be positioned in the inner space of the second housing **32** so as to overlap with at least a portion of the display area of the second display module **35** when viewed from above the second cover **322**. In this case, the first camera module **405** or the position of the first camera module **405** may be visually distinguished (or exposed) or invisible. The first camera module **405** may be positioned, for example, on the rear surface of the second display module **35** or below or beneath the second display module **35**, and the first camera module **405** or the position of the first camera module **405** may be visually distinguished (or exposed) or invisible. In various embodiments, the first camera module **405** may be positioned in alignment with or at least partially inserted into a recess provided on the rear surface of the second display module **35**. The first camera module **405** may include, for example, a hidden display rear camera (e.g., an under display camera (UDC)). A portion of the second display module **35** that at least partially overlaps with the first camera module **405** may include a different pixel structure and/or wiring structure from other portions (not shown separately). For example, a portion of the second display module **25** that at least partially overlaps with the first camera module **405** may have a different pixel density (e.g., number of pixels per unit area) from other portions. The pixel structure and/or wiring structure formed in a portion of the second display module **35** that at least partially overlaps with the first camera module **405** may reduce light loss between the outside and the first camera module **405**. For another example, a plurality of pixels may not be disposed in a portion of the second display module **35** that at least partially overlaps with the first camera module **405**.

[0118] According to an embodiment, the second camera module **406**, the third camera module **407**, or the fourth camera module **408** may be positioned in the inner space of the first housing **31** corresponding to the first cover **312**. The first housing **31** or the first cover **312** may include a camera cover portion (e.g., a camera decoration portion) **410** disposed corresponding to the second camera module **406**, the third camera module **407**, and the fourth camera module **408**. The camera cover portion **410** may include a camera hole (or light transmitting area) provided corresponding to the second camera module **406**, a camera hole (or light transmitting area) provided corresponding to the third camera module **407**, and a camera hole (or light transmitting area) provided corresponding to the fourth camera module **408**. The number or location of camera modules provided in response to the first cover **312** is not limited to the illustrated example and may vary.

[0119] According to an embodiment, the second camera module **406**, the third camera module **407**, and the fourth camera module **408** may have different properties (e.g., angles of view) or functions. The second camera module **406**, the third camera module **407**, and the fourth camera module **408** may provide different angles of view (or lenses with different angles of view), and the foldable electronic device **3** may selectively use at least one camera module on the basis of a user's selection regarding the angle of view. Any one of the second camera module **406**, the third camera module **407**, and the fourth camera module **408** may include a wide-angle camera module, a telephoto camera module, a color camera module, a monochrome camera module, or an infrared (IR) camera

(e.g., a time of flight (TOF) camera, a structured light camera) module. In various embodiments, the IR camera module may also operate as at least a portion of a sensor module.

[0120] According to an embodiment, the light-emitting module **409** may be positioned in the inner space of the first housing **31** corresponding to a flash hole (or light transmitting area) provided in the camera cover portion **310** of the first cover **312**. The light-emitting module **409** may include a light source for the second camera module **406**, the third camera module **407**, and/or the fourth camera module **408**. The light-emitting module **409** may include various light sources such as a light emitting diode (LED) or a xenon lamp.

[0121] According to various embodiments, any one of the one or more light-emitting modules (e.g., an LED, an IR LED, or a xenon lamp) (not shown separately) may be configured to provide status information of the foldable electronic device **3** in the form of light. In various embodiments, any one of the one or more light-emitting modules (not shown separately) may be positioned corresponding to the first front surface and may provide a light source that is linked to the operation of the first camera module **405**.

[0122] According to an embodiment, one of the one or more connecting terminals may include a first connector (or connector module or interface terminal) (not illustrated separately) positioned inside the foldable electronic device **3** corresponding to a first connector hole **411** formed on an exterior (e.g., a first side surface) of the foldable electronic device **3**. The first connector may be, for example, a universal serial bus (USB) connector or a high definition multimedia interface (HDMI) connector. The foldable electronic device **3** may transmit and/or receive data and/or power with an external electronic device electrically connected to the first connector. The positions or numbers of the first connector and the connector holes **411** corresponding to the first connector are not limited to the illustrated examples and may vary.

[0123] According to various embodiments, one of the one or more connecting terminals may include a second connector (or a second connector module or a second interface terminal) (not shown separately) positioned inside the foldable electronic device **3** corresponding to a second connector hole (not shown separately) formed on an exterior surface (e.g., a first side surface or a second side surface) of the foldable electronic device **3**. For example, an external storage medium (e.g., a subscriber identity module (SIM) card or a memory card (e.g., a secure digital memory (SD) card)) may be connected to the second connector.

[0124] According to an embodiment, one of the one or more key input modules may include a first key **412** positioned on an exterior (e.g., a first side surface) of the foldable electronic device **3** and a key signal generating unit positioned inside the foldable electronic device **3** corresponding to the first key **412**. Another one of the one or more key input modules may include a second key **413** positioned on the exterior of the foldable electronic device **3** and a key signal generating unit positioned inside the foldable electronic device **3** corresponding to the second key **413**. The positions or number of the keys are not limited to the illustrated examples and may vary.

[0125] According to an embodiment, the antenna module **5** may be positioned inside the foldable electronic device **3** corresponding to the first cover **312**. The foldable electronic device **3** may transmit an electromagnetic signal (or a radio signal, an RF signal, a radio wave, or an electromagnetic wave) to the outside of the foldable electronic device **3** through the antenna module **5**. The electromagnetic signal from the antenna module **5** may penetrate the first cover **312** and proceed to the outside of the foldable electronic device **3**. The foldable electronic device **3** may receive an electromagnetic signal from the outside of the foldable electronic device **3** through the antenna module **5**. The electromagnetic signal from the outside of the foldable electronic device **3** may penetrate the first cover **312** and reach the antenna module **5**.

[0126] According to various embodiments, the antenna module **5** may be disposed at various other locations facing the foldable housing **30** (not shown separately).

[0127] According to various embodiments, the foldable electronic device **3** may include a detachable pen input device (e.g., an electronic pen, a digital pen, or a stylus pen) (not shown

separately). The pen input device may, for example, be implemented so as to be inserted into an inner space of the first housing **31** or the second housing **32**. As another example, the pen input device may be detachably attached to a hinge housing **33**. The hinge housing **33** may include a recess, and the pen input device may be fitted into the recess.

[0128] The foldable electronic device **3** may further include various components depending on its provision form. These components are not all listed because they are modified in various ways depending on the convergence trend of the foldable electronic device **3**, but components equivalent to the above-mentioned components may be additionally included in the foldable electronic device **3**. In various embodiments, specific components may be excluded from the above-mentioned components or replaced with other components depending on their provision form.

[0129] According to various embodiments, the disclosure illustrates, but is not limited to, a foldable electronic device **3**. The scope of various embodiments of the disclosure may be applied to bar type or plate type electronic devices, slidable electronic devices, stretchable electronic devices, and/or rollable electronic devices.

[0130] FIGS. **5** and **6** are diagrams illustrating an antenna module **5** according to various embodiments of the disclosure. FIGS. **7** and **8** are diagrams illustrating beam patterns generated from an antenna module **5** according to various embodiments of the disclosure.

[0131] Referring to FIGS. **5**, **6**, **7**, and **8**, the antenna module **5** may include an antenna structure **51**, a communication circuit (or wireless communication circuit) **54**, a power management circuit **55**, a first dielectric **61**, a second dielectric **62**, a third dielectric **63**, and/or a fourth dielectric **64**.

[0132] According to various embodiments, the antenna module **5** may be the third antenna module **246** of FIG. **2**.

[0133] According to an embodiment, the antenna structure **51** may include a printed circuit board **510**. The printed circuit board **510** may include a first surface **511** and a second surface **512** facing in an opposite direction from the first surface **511**. The first surface **511** and the second surface **512** may be, for example, substantially parallel.

[0134] According to an embodiment, the first surface **511** of the printed circuit board **510** may face the first cover **312** (see FIG. **3**). A second surface **512** of the printed circuit board **510** may face the first display area {circle around (1)} of the first display module **34**. According to an embodiment, the printed circuit board **510** may include a plurality of pattern layers (not shown separately) including conductive patterns, and a plurality of non-conductive layers (e.g., insulating layers) (not shown separately) stacked in alternation with the plurality of pattern layers.

[0135] According to an embodiment, the printed circuit board **510** may include a plurality of conductive vias (not shown separately). A conductive via may be a conductive hole provided to allow for disposition of a connecting wire for electrically connecting conductive patterns of different pattern layers. The conductive via may, for example, include a plated through hole (PTH), a laser through hole (LVH), a buried through hole (BVH), or a stacked via. At least one conductive via may, for example, electrically connect signal line patterns (e.g., patterns of conductive lines utilized as electrical paths) included in different pattern layers as a portion of a signal line. At least one conductive via may, for example, electrically connect ground planes (or ground areas or ground patterns) included in different pattern layers.

[0136] According to an embodiment, the printed circuit board **510** may include an antenna array **52**. For example, the antenna array **52** may be implemented as some conductive patterns among a plurality of pattern layers included in the printed circuit board **510**. The antenna array **52** may be disposed on the first surface **511** or may be disposed within the printed circuit board **510** closer to the first surface **511** than to the second surface **512**. The antenna array **52** may include a plurality of antenna elements **521**, **522**, **523**, **524**, and **525**. The plurality of antenna elements **521**, **522**, **523**, and **524** may, for example, be the antenna **248** of FIG. **2**.

[0137] According to an embodiment, the plurality of antenna elements **521**, **522**, **523**, and **524** may have substantially the same shape and may be disposed at a predetermined interval. The plurality of

antenna elements **521**, **522**, **523**, and **524** may be disposed in a sixth direction **501** (e.g., the y-axis direction) when viewed from above the first surface **511** (e.g., when viewed in the +z-axis direction). The sixth direction **501** may, for example, be substantially parallel to the center line A of the foldable electronic device **3** (see FIG. **3**).

[0138] According to various embodiments, the plurality of antenna elements **521**, **522**, **523**, and **524** may be disposed in a seventh direction **502** (e.g., the x-axis direction) that is perpendicular to the sixth direction **501** when viewed from above the first surface **511**.

[0139] According to an embodiment, when viewed from above the first surface **511** (e.g., when viewed in the +z-axis direction), the plurality of antenna elements **521**, **522**, **523**, and **524** may be substantially square. For example, the first antenna element **521** may include a first edge **E1**, a second edge **E2**, a third edge **E3**, and a fourth edge **E4** when viewed from above the first surface **511**. When viewed from above the first surface **511**, the first edge **E1** and the second edge **E2** may be spaced apart from each other in a sixth direction **501** and may be substantially parallel to each other. When viewed from above the first surface **511**, the third edge **E3** and the fourth edge **E4** may be spaced apart from each other in the seventh direction **502** and may be substantially parallel to each other. The remaining antenna elements **522**, **523**, and **524** may have substantially the same shape as the first antenna element **521**. The shapes of the plurality of antenna elements **521**, **522**, **523**, and **524** are not limited thereto and may vary. The plurality of antenna elements **521**, **522**, **523**, and **524** may, for example, be provided in various polygonal shapes, such as a circle, an ellipse, or a triangle (not shown separately).

[0140] According to various embodiments, the number or location of antenna elements included in the antenna array **52** is not limited to the illustrated embodiment and may vary (not illustrated separately).

[0141] According to various embodiments, the number or location of the antenna arrays **52** is not limited to the illustrated embodiment and may vary (not illustrated separately).

[0142] According to an embodiment, the printed circuit board **510** may include a ground plane **53** that acts as an antenna ground. The ground plane **53** may contribute to securing antenna radiation performance (or radio transmission/reception performance or communication performance) and/or coverage with respect to the plurality of antenna elements **521**, **522**, **523**, and **524**. The ground plane **53** may reduce electromagnetic interference (EMI) (or signal loss) with respect to the plurality of antenna elements **521**, **522**, **523**, and **524**.

[0143] According to an embodiment, the ground plane **53** may be disposed closer to the second surface **512** of the printed circuit board **510** than the antenna array **52**. When viewed from above the first surface **511** of the printed circuit board **510** (e.g., when viewed in the +z-axis direction), the antenna array **52** may at least partially overlap with the ground plane **53**. For example, the antenna array **52** may be included in a first pattern layer (not shown separately) of the printed circuit board **510**, and the ground plane **53** may be included in a second pattern layer (not shown separately) closer to the second surface **512** than the first pattern layer.

[0144] According to an embodiment, the antenna structure **51** may operate as a patch antenna. When the communication circuit **54** provides (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the plurality of antenna elements **521**, **522**, **523**, and **524**, the antenna structure **51** may operate as a resonator (e.g., a patch antenna) because the plurality of antenna elements **521**, **522**, **523**, and **524** act as radiating elements while resonating with the ground plane **53**. The plurality of antenna elements **521**, **522**, **523**, and **524** may be interpreted as a plurality of patches.

[0145] According to an embodiment, the plurality of antenna elements **521**, **522**, **523**, and **524** may be included in a single pattern layer (not shown separately) among the plurality of pattern layers included in the printed circuit board **510** (e.g., a single layer patch structure). For example, the plurality of antenna elements **521**, **522**, **523**, and **524** may be directly fed from the communication circuit **54**.

[0146] According to various embodiments, the printed circuit board **510** may further include a plurality of dummy antenna elements (e.g., dummy patches) (not shown separately) (e.g., a stacked patch structure). The plurality of dummy antenna elements may be positioned to overlap with the plurality of antenna elements **521**, **522**, **523**, and **524** one-to-one when viewed from above the first surface **511** and may be electrically floating. One pattern layer including the plurality of dummy antenna elements may be positioned closer to the first surface **511** than another pattern layer including the plurality of antenna elements **521**, **522**, **523**, and **524**. The communication circuit **54** may provide (or feeding) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to a plurality of antenna elements **521**, **522**, **523**, and **524**, and the dummy antenna elements may be indirectly fed from the plurality of antenna elements **521**, **522**, **523**, and **524**. The plurality of antenna elements **521**, **522**, **523**, and **524** may be interpreted as a plurality of feeding antenna elements. The plurality of dummy antenna elements may be electromagnetically coupled with the plurality of antenna elements **521**, **522**, **523**, and **524** to act as part of a radiator and adjust radiation features. The plurality of dummy antenna elements may, for example, shift a resonant frequency of the antenna module **5** to a specified frequency or by a specified amount. A plurality of dummy antenna elements may, for example, expand the bandwidth through which an electromagnetic signal can be transmitted or received through the antenna module **5** or form different frequency bands (e.g., multi-bands). A plurality of dummy antenna elements may, for example, improve antenna performance by reducing electromagnetic noise.

[0147] According to various embodiments, the printed circuit board **510** may further include a plurality of feeding antenna elements (not shown) that at least partially overlap with the plurality of antenna elements **521**, **522**, **523**, and **524** and are physically separated from the plurality of antenna elements **521**, **522**, **523**, and **524** when viewed from above the first surface **511** (e.g., a stacked patch structure). A pattern layer including the plurality of antenna elements **521**, **522**, **523**, and **524** may be positioned closer to the first surface **511** than a pattern layer including the plurality of feeding antenna elements. A communication circuit **54** may provide (or feeding) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to a plurality of feeding antenna elements, and the plurality of antenna elements **521**, **522**, **523**, and **524** may be indirectly fed from the plurality of feeding antenna elements. The plurality of antenna elements **521**, **522**, **523**, and **524** may be interpreted as a plurality of dummy antenna elements.

[0148] According to an embodiment, the communication circuit **54** may be disposed on the second surface **512** of the printed circuit board **510** through a conductive adhesive material (or conductive bonding material) such as solder. The communication circuit **54** may be electrically connected to the plurality of antenna elements **521**, **522**, **523**, and **524** through wires (or electrical paths) included in the printed circuit board **510**.

[0149] According to an embodiment, the communication circuit **54** may include a radio frequency integrated circuit (RFIC). The communication circuit **54** may be, for example, the third RFIC **226** of FIG. **2**.

[0150] According to various embodiments, the communication circuit **54** may be disposed on a printed circuit board different from the printed circuit board **510** (e.g., a printed circuit board on which components such as the processor **120**, the memory **130**, or the communication module **190** of FIG. **1** are disposed). For example, the communication circuit **54** may be omitted from the antenna module **5**.

[0151] According to an embodiment, the communication circuit **54** may up-convert or down-convert the frequency of a signal being transmitted or received. For example, the communication circuit **54** may receive an intermediate frequency (IF) signal from a wireless communication circuit (e.g., the wireless communication module **192** of FIG. **1**) disposed on another printed circuit board, and up-convert the received IF signal to a radio frequency (RF) signal (e.g., a millimeter wave). As another example, the communication circuit **54** may down-convert an RF signal received through the antenna array **52** to an IF signal, and the IF signal may be provided to a wireless

communication circuit (e.g., the wireless communication module **192** of FIG. **1**) disposed on another printed circuit board.

[0152] According to an embodiment, the communication circuit **54** may transmit and/or receive signals of substantially the same frequency band (or frequency) through a plurality of antenna elements **521**, **522**, **523**, and **524**.

[0153] According to an embodiment, the communication circuit **54** may transmit and/or receive signals in at least a portion of a frequency band from about 3 GHz to about 100 GHz through a plurality of antenna elements **521**, **522**, **523**, and **524**.

[0154] According to an embodiment, the power management circuit **55** may be disposed on the second surface **512** of the printed circuit board **510** through a conductive adhesive material (or conductive bonding material), such as solder. The power management circuit **55** may be electrically connected to the communication circuit **54** or various other components (e.g., passive components) disposed on the printed circuit board **510** through wires (or electrical paths) included in the printed circuit board **510**.

[0155] According to an embodiment, the power management circuit **55** may include a power management integrated circuit (PMIC).

[0156] According to various embodiments, the power management circuit **55** may be disposed on a printed circuit board different from the printed circuit board **510** (e.g., a printed circuit board on which components such as the processor **120**, the memory **130**, or the communication module **190** of FIG. **1** are disposed). For example, the power management circuit **55** may be omitted from the antenna module **5**.

[0157] According to an embodiment, the antenna module **5** may be electrically connected to another printed circuit board (e.g., a printed circuit board on which components such as the processor **120**, the memory **130**, or the communication module **190** of FIG. **1** are disposed) through an electrical connection member (not shown separately) such as a flexible printed circuit board. For example, one end of the electrical connection member (not shown separately) may be electrically connected to a connector (e.g., an FPCB connector) (not shown separately) disposed on the second surface **512**, and the other end of the electrical connection member may be electrically connected to a connector disposed on another printed circuit board. As another example, one end of the electrical connection member (not shown separately) may be electrically connected to conductive terminals (e.g., lands or copper pads) provided on the second surface **512** through a conductive adhesive material (or a conductive bonding material) such as solder.

[0158] According to various embodiments, the antenna module **5** may further include a shielding member (or electromagnetic shielding member) (not shown) surrounding at least one of the communication circuit **54** and/or the power management circuit **55**. The shielding member may reduce electromagnetic interference (e.g., electromagnetic noise) regarding the communication circuit **54** and/or the power management circuit **55**. The shielding member may include a conductive member, such as a shield can, for example. As another example, the shielding member may include a protective member, such as a urethane resin, and a conductive paint, such as an electromagnetic interference (EMI) paint, applied to an outer surface of the protective member. In various embodiments, the shielding member may be provided as various shielding sheets.

[0159] According to various embodiments, the antenna module **5** may further include a frequency tuning circuit (not shown) disposed on a printed circuit board **510**. The frequency tuning circuit may include, for example, a tuner or a passive element. The frequency tuning circuit may shift the impedance matching, or resonant frequency, to a specified frequency or by a specified amount.

[0160] According to an embodiment, the first dielectric **61**, the second dielectric **62**, the third dielectric **63**, and/or the fourth dielectric **64** may be disposed on or coupled to a first surface **511** of a printed circuit board **510**. For example, the first dielectric **61** may be positioned corresponding to the first antenna element **521**. When viewed from above the first surface **511**, the first dielectric **61** may overlap with the first antenna element **521**. For example, the second dielectric **62** may be

positioned corresponding to the second antenna element **522**. When viewed from above the first surface **511**, the second dielectric **62** may overlap with the second antenna element **522**. For example, the third dielectric **63** may be positioned corresponding to the third antenna element **523**. When viewed from above the first surface **511**, the third dielectric **63** may overlap with the third antenna element **523**. For example, the fourth dielectric **64** may be positioned corresponding to the fourth antenna element **524**. When viewed from above the first surface **511**, the fourth dielectric **64** may overlap with the fourth antenna element **524**.

[0161] According to an embodiment, the first dielectric **61**, the second dielectric **62**, the third dielectric **63**, and the fourth dielectric **64** may be provided in substantially the same form.

[0162] According to an embodiment, the first dielectric **61**, the second dielectric **62**, the third dielectric **63**, and the fourth dielectric **64** may include non-conductive materials. With respect to a frequency (e.g., a resonant frequency) of an electromagnetic signal transmitted and/or received through the antenna module **5**, the first dielectric **61**, the second dielectric **62**, the third dielectric **63**, and the fourth dielectric **64** may have a relative permittivity that is greater than that of a conductor having a relative permittivity being substantially 1.

[0163] According to an embodiment, the antenna module **5** may include a first electrical path (or a first conductive path) (not shown) for transmitting an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the first dielectric **61**. A first end of the first electrical path may be electrically connected to a communication circuit **54**. A second end of the first electrical path may be in physical contact with or directly connected to the first dielectric **61**. A portion of the first dielectric **61** through which the electromagnetic signal is transmitted from the second end of the first electrical path may be defined or interpreted as a first feeding portion **FP1**. In some embodiments, the first electrical path, or the second end of the first electrical path, may be defined or interpreted as a 'first feeding portion', and the portion of the first dielectric **61** through which an electromagnetic signal is transmitted from the second end of the first electrical path may be defined or interpreted as a 'first feeding area' or a 'first feeding point'. The provision of an electromagnetic signal by a communication circuit (e.g., the communication circuit **54** of FIG. **6**) to a first feeding portion **FP1** (or a first electrical path, a first feeding area, or a first feeding point) may be defined or interpreted as a 'first feeding'.

[0164] According to various embodiments, the first electrical path may be interpreted as a first transmission line or a first feeding line along which an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) is transmitted.

[0165] According to an embodiment, when the communication circuit **54** transmits (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) through the first electrical path, the first dielectric **61** may operate as a first dielectric resonator (or a first dielectric resonator antenna (DRA)). When the communication circuit **54** transmits (or feedings) an electromagnetic signal through the first electrical path, the first dielectric resonator may be provided because of a discontinuity between the first dielectric **61** and a medium in contact with (or surrounding) the first dielectric **61**. The medium in contact with (or surrounding) the first dielectric **61** may include, for example, air. When the first dielectric **61** contacts the first cover **312** (see FIG. **3**), the medium in contact with the first dielectric **61** (or surrounding the first dielectric **61**) may include a first cover **312**.

[0166] According to various embodiments, the second end of the first electrical path may be indirectly connected to the first dielectric **61**. For example, another dielectric (not shown separately) may be disposed between the second end of the first electrical path and the first dielectric **61**, and the first dielectric **61** may indirectly receive an electromagnetic signal from the second end through the other dielectric. A combination of the first dielectric **61** and the other dielectric may be defined or interpreted as a dielectric resonator, and the first feeding portion **FP1** may be included in the other dielectric.

[0167] According to an embodiment, the first electrical path may be included in a printed circuit

board **510**.

[0168] According to various embodiments, the first electrical path may include a first partial electrical path (or first partial conductive path) (not shown separately), and a second partial electrical path (or second partial conductive path) (not shown separately) electrically connected to the first partial electrical path. For example, the first partial electrical path may be included in the printed circuit board **510**. For example, the second partial electrical path may be included in a separate structure (or member) (not shown separately) disposed on or coupled to the printed circuit board **510** (e.g., the first surface **511** of the printed circuit board **510**).

[0169] According to an embodiment, the first electrical path may be physically separated from the plurality of antenna elements **521**, **522**, **523**, and **524**.

[0170] According to an embodiment, the first antenna element **521** may include an opening **5211** corresponding to a first electrical path. The first electrical path may penetrate the opening **5211**.

[0171] According to an embodiment, the opening **5211** may be provided at the center of the first antenna element **521**. When viewed from above the first surface **511**, a distance at which the opening **5211** is spaced apart from the first edge **E1** in the sixth direction **501** and a distance at which the opening **5211** is spaced apart from the second edge **E2** in the sixth direction **501** may be substantially the same. When viewed from above the first surface **511**, a distance at which the opening **5211** is spaced apart from the third edge **E3** in the seventh direction **502** and a distance at which the opening **5211** is spaced apart from the fourth edge **E4** in the seventh direction **502** may be substantially the same.

[0172] According to an embodiment, when viewed from above the first surface **511**, a distance at which the first feeding portion **FP1** is spaced apart from the first edge **E1** in the sixth direction **501** and a distance at which the first feeding portion **FP1** is spaced apart from the second edge **E2** in the sixth direction **501** may be substantially the same. When viewed from above the first surface **511**, a distance at which the first feeding portion **FP1** is spaced apart from the third edge **E3** in the seventh direction **502** and a distance at which the first feeding portion **FP1** is spaced apart from the fourth edge **E4** in the seventh direction **502** may be substantially the same.

[0173] According to an embodiment, the printed circuit board **510** may include a second electrical path (not shown) for transmitting an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the first antenna element **521**. The second electrical path may electrically connect the first antenna element **521** and the communication circuit **54**. The portion of the first antenna element **521** through which the electromagnetic signal is transmitted from the second electrical path may be defined or interpreted as a second feeding portion **FP2**. In some embodiments, the second electrical path, or the portion of the second electrical path connected to the first antenna element **521**, may be defined or interpreted as a 'second feeding portion', and the portion of the first antenna element **521** through which the electromagnetic signal is transmitted from the second electrical path may be defined or interpreted as a 'second feeding area' or a 'second feeding point'. The provision of an electromagnetic signal by a communication circuit (e.g., the communication circuit **54** of FIG. **6**) to a second feeding portion **FP2** (or a second electrical path, a second feeding area, or a second feeding point) may be defined or interpreted as a 'second feeding'.

[0174] According to various embodiments, the second electrical path may be interpreted as a second transmission line or a second feeding line along which an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) is transmitted.

[0175] According to an embodiment, when viewed from above the first surface **511**, the second feeding portion **FP2** may be positioned between the opening **5211** (or the first feeding portion **FP1**) and the first edge **E1**.

[0176] According to an embodiment, when viewed from above the first surface **511**, the second feeding portion **FP2** may be positioned closer to the first edge **E1** than the opening **5211** (or the first feeding portion **FP1**).

[0177] According to various embodiments, the second feeding portion FP2 may be substantially positioned at the first edge E1 (not shown separately).

[0178] According to an embodiment, when the communication circuit 54 provides (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the second feeding portion FP2 through the second electrical path, a signal having a first linear polarization may be radiated from the first antenna element 521. When the communication circuit 54 provides an electromagnetic signal to the feeding portion FP2, a first linear polarization having a first polarization direction may be radiated by a current path (or a distribution of surface current) on the first antenna element 521. The first linear polarization may be an electromagnetic wave that propagates while oscillating in the first polarization direction. The first linear polarization may be interpreted as a polarity direction of an electric field with respect to a propagation direction (or transmission direction) of an electromagnetic wave radiated (or transmitted) from the first antenna element 521. The first polarization direction is a direction in which the electric field oscillates (or a direction parallel to a vector of the electric field) and may be perpendicular to the propagation direction of the electromagnetic wave. The propagation direction of the electromagnetic wave may be a direction in which a plane of the first antenna element 521 in the form of a flat plate (or plane) faces (for example, a -z-axis direction in which a first surface 511 of a printed circuit board 510 faces).

[0179] According to an embodiment, the first polarization direction of the first linear polarization radiated from the first antenna element 521 when fed by the second feeding portion FP2 may be substantially in the sixth direction 501.

[0180] According to an embodiment, the printed circuit board 510 may include a third electrical path (not shown) for transmitting an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the first antenna element 521. The third electrical path may electrically connect the first antenna element 521 and the communication circuit 54. A portion of the first antenna element 521 through which the electromagnetic signal is transmitted from the third electrical path may be defined or interpreted as a third feeding portion FP3. In some embodiments, the third electrical path, or a portion of the third electrical path connected to the first antenna element 521, may be defined or interpreted as a 'third feeding portion', and a portion of the first antenna element 521 through which the electromagnetic signal is transmitted from the third electrical path may be defined or interpreted as a 'third feeding area' or a 'third feeding point'. The provision of an electromagnetic signal by a communication circuit (e.g., the communication circuit 54 of FIG. 6) to a third feeding portion FP3 (or a third electrical path, a third feeding area, or a third feeding point) may be defined or interpreted as a 'third feeding'.

[0181] According to various embodiments, the third electrical path may be interpreted as a third transmission line or a third feeding line along which an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) is transmitted.

[0182] According to an embodiment, when viewed from above the first surface 511, the third feeding portion FP3 may be positioned between the opening 5211 (or the first feeding portion FP1) and the third edge E1. When viewed from above the first surface 511, the second feeding portion FP2 and the third feeding portion FP3 may form, for example, a substantially 90 degree angle with respect to the opening 5211 (or the first feeding portion FP1).

[0183] According to an embodiment, when viewed from above the first surface 511, the third feeding portion FP3 may be positioned closer to the third edge E3 than the opening 5211 (or the first feeding portion FP1).

[0184] According to various embodiments, the third feeding portion FP3 may be substantially located at the third edge E3 (not shown separately).

[0185] According to an embodiment, when viewed from above the first surface 511, the distance at which the third feeding portion FP3 is spaced from the third edge E3 in the seventh direction 502 may be substantially equal to the distance at which the second feeding portion FP2 is spaced from

the first edge **E1** in the sixth direction **501**.

[0186] According to an embodiment, when viewed from above the first surface **511**, the distance at which the third feeding portion **FP3** is spaced from the opening **5211** in the seventh direction **502** may be substantially equal to the distance at which the second feeding portion **FP2** is spaced from the opening **5211** in the sixth direction **501**.

[0187] According to an embodiment, when the communication circuit **54** provides (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the third feeding portion **FP3** through the third electrical path, a signal having a second linear polarization may be radiated from the first antenna element **521**. When the communication circuit **54** provides an electromagnetic signal to the third feeding portion **FP3**, a second linear polarization having a second polarization direction (or oscillating in the second polarization direction) may be radiated by the current path (or distribution of surface current) on the first antenna element **521**. The second polarization direction may be perpendicular to the propagation direction of the electromagnetic wave (e.g., $-z$ axis direction).

[0188] According to an embodiment, the second polarization direction of the second linear polarization radiated from the first antenna element **521** when fed by the third feeding portion **FP3** may be substantially in the seventh direction **502**.

[0189] According to various embodiments, a single feeding portion may be provided to replace the second feeding portion **FP2** and the third feeding portion **FP3**. The single feeding portion may be substantially located, for example, at a first corner (e.g., a first vertex) where the first edge **E1** and the third edge **E3** meet, a second corner (e.g., a second vertex) where the second edge **E2** and the third edge **E3** meet, a third corner (e.g., a third vertex) where the second edge **E2** and the fourth edge **E4** meet, or a fourth corner (e.g., a fourth vertex) where the first edge **E1** and the fourth edge **E4** meet, when viewed from above the first surface **511**. For another example, a single feeding portion may be located between the opening **5211** (or the first feeding portion **FP1**) and the first corner, between the opening **5211** (or the first feeding portion **FP1**) and the second corner, between the opening **5211** (or the first feeding portion **FP1**) and the third corner, or between the opening **5211** (or the first feeding portion **FP1**) and the fourth corner, when viewed from above the first surface **511**. When the communication circuit **54** provides (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the feeding portion, the linear polarization radiated from the first antenna element **521** may have a polarization direction substantially parallel to a diagonal direction that forms a substantially 45 degree angle with respect to the sixth direction **501** or the seventh direction **502** when viewed from above the first surface **511**. The linear polarization having a diagonal polarization direction may be interpreted as a composite of a first polarization component having a first polarization direction parallel to the sixth direction **501** and a second polarization component parallel to the seventh direction **502**.

[0190] According to various embodiments, the shape of the first antenna element **521** and the position of the feeding portion are not limited to the illustrated example, and may be variously provided so as to radiate linear polarization corresponding to at least one selected or designated frequency band. The linear polarization may be interpreted as a composite of a first polarization component and a second polarization component each having a polarization direction that is perpendicular to each other.

[0191] According to various embodiments, the shape of the first antenna element **521** and the position of the feeding portion are not limited to the illustrated example, and may be variously provided so as to be able to radiate a circular polarization corresponding to at least one selected or designated frequency band. The circular polarization may be interpreted as a composite of a first polarization component and a second polarization component having polarization directions that are perpendicular to each other. When the phase difference between the first polarization component and the second polarization component is substantially 90 degrees, the composite of the first polarization component and the second polarization component may provide a composite wave

having the feature of a circular polarization where an axial ratio is substantially close to 1.

[0192] According to various embodiments, the shape of the first dielectric **61** and the position or number of the first feeding portions **FP1** are not limited to the illustrated example and may vary (not illustrated separately). For example, the antenna module **5** may include a plurality of first electrical paths physically connected to a plurality of first feeding portions (not illustrated separately) of the first dielectric **61**.

[0193] According to various embodiments, the shape or position of the opening **5211** provided in the first antenna element **521** is not limited to the illustrated example and may vary (not illustrated separately). For example, the opening **5211** may be provided in a polygonal shape such as an oval or a triangle, not limited to the illustrated circular shape, when viewed from above the first surface **511**. For another example, the opening **5211** may be replaced by a notch provided in the first edge **E1**, the second edge **E2**, the third edge **E3**, or the fourth edge **E4**.

[0194] According to various embodiments, the opening **5211** of the first antenna element **521** may be omitted. For example, the first feeding portion **FP1** of the first dielectric **61** may be positioned in a portion of the first dielectric **61** that does not overlap with the first antenna element **521** when viewed from above the first surface **511**.

[0195] According to an embodiment, a first radiating portion (or a first resonator or a first resonant portion) **710** provided by a combination of a first dielectric resonator (or a first dielectric resonant portion) including a first dielectric **61** and a resonator (hereinafter, a 'first patch resonator' or 'a first patch resonant portion') including a first antenna element **521** may improve (e.g., expand) the coverage of the antenna module **5** or improve the antenna radiation performance of the antenna module **5** compared to a comparative example in which the first dielectric resonator is omitted.

[0196] According to an embodiment, the foldable electronic device **3** (see FIG. **3**) may include a beamforming system (or a beamforming circuit) for the antenna module **5**. The beamforming system may enable receiving a stronger signal in a desired direction, transmitting a signal in a desired direction, or preventing receiving a signal from an undesired direction. The beamforming system included in the communication circuit **54** may control the phase or amplitude of an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to adjust a beam pattern (e.g., shape and direction of the beam).

[0197] According to an embodiment, a processor (e.g., the processor **120** of FIG. **1**) may determine a phase of an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) provided to the first feeding portion **FP1**, the second feeding portion **FP2**, or the third feeding portion **FP3** according to a frequency (or a frequency band) used by an application (or a program) on the basis of codebook information about beamforming stored in a memory (e.g., the memory **130** of FIG. **1**). The communication circuit **54** of the antenna module **5** may adjust the phase of an electromagnetic signal provided to the first feeding portion **FP1**, the second feeding portion **FP2**, or the third feeding portion **FP3** under the control of the processor. The communication circuit **54** of the antenna module **5** may efficiently control (e.g., allocate or dispose) a plurality of beams through a plurality of radiating portions (or resonators) included in the antenna module **5** under the control of the processor.

[0198] According to an embodiment, a second radiation field (or a second electromagnetic field) may be generated from the first patch resonator of the first radiating portion **710** when the communication circuit **54** has a second feeding that provides an electromagnetic signal (or a radio signal, RF signal, or radiated current) to the second feeding portion **FP2**.

[0199] According to an embodiment, the second radiation field generated during the second feeding to the second feeding portion **FP2** may provide a first beam pattern **711** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a first direction {circle around (1)} substantially directed toward the first surface **511**. The first direction {circle around (1)} may be substantially perpendicular to the sixth direction **501** and the seventh direction **502**.

[0200] According to an embodiment, the second radiation field generated during the second feeding to the second feeding portion FP2 and the first beam pattern **711** provided by the second radiation field, for example, may correspond to a first linear polarization having a substantially first polarization direction (e.g., a substantially sixth direction **501**).

[0201] According to an embodiment, when the communication circuit **54** has a first feeding providing an electromagnetic signal (or a radio signal, RF signal, or radiated current) to a first feeding portion FP1, and a second feeding providing an electromagnetic signal to a second feeding portion FP2, a first radiation field (or a first electromagnetic field) may be generated from a first dielectric resonator of the first radiating portion **710**, and a second radiation field (or a second electromagnetic field) may be generated from a first patch resonator of the first radiating portion **710**. Because of interference (e.g., constructive interference and/or destructive interference) between the first radiation field and the second radiation field, a second beam pattern **712** or a third beam pattern **713** may be provided, which is capable of radiating or concentrating a relatively more electromagnetic wave energy (or wave energy) in a direction different from the first direction {circle around (1)} (or tilted with respect to the first direction {circle around (1)}).

[0202] According to an embodiment, the communication circuit **54** may adjust the phase difference of the electromagnetic signal provided to the first feeding portion FP1 and the electromagnetic signal provided to the second feeding portion FP2. The interference between the first radiation field generated from the first dielectric resonator of the first radiating portion **710** and the second radiation field generated from the first patch resonator of the first radiating portion **710** may provide a second beam pattern **712** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a second direction {circle around (2)} different from the first direction {circle around (1)} because of the phase difference between the electromagnetic signals provided to the first feeding portion FP1 and the electromagnetic signals provided to the second feeding portion FP2. The interference between the first radiation field generated from the first dielectric resonator of the first radiating portion **710** and the second radiation field generated from the first patch resonator of the first radiating portion **710** may provide a third beam pattern **713** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a third direction {circle around (3)} different from the first direction {circle around (1)} and the second direction {circle around (2)} because of the phase difference between the electromagnetic signals provided to the first feeding portion FP1 and the electromagnetic signals provided to the second feeding portion FP2.

[0203] According to an embodiment, the second direction {circle around (2)} corresponding to the second beam pattern **712** and the third direction {circle around (3)} corresponding to the third beam pattern **713** may be substantially perpendicular to the seventh direction **502** (e.g., the y-axis direction) and may form an acute or obtuse angle with the first direction {circle around (1)} (e.g., the -z-axis direction). The second direction {circle around (2)} and the third direction {circle around (3)} may be provided substantially symmetrically when viewed from the seventh direction **502**.

[0204] According to an embodiment, for the second beam pattern **712**, the phase difference between the electromagnetic signal provided to the first feeding portion FP1 and the electromagnetic signal provided to the second feeding portion FP2 may be substantially -90° . For the third beam pattern **713**, the phase difference between the electromagnetic signal provided to the first feeding portion FP1 and the electromagnetic signal provided to the second feeding portion FP2 may be substantially $+90^\circ$.

[0205] According to an embodiment, the second beam pattern **712** or third beam pattern **713** radiated during the first feeding to the first feeding portion FP1 and the second feeding to the second feeding portion FP2 may correspond to a first linear polarization having a substantially first polarization direction (e.g., a substantially sixth direction **501**). The first radiating portion **710** including the first dielectric resonator and the first patch resonator may extend coverage (or beam

coverage), compared to a comparative example in which the first dielectric resonator is omitted. The first radiating portion **710** including the first dielectric resonator and the first patch resonator may concentrate electromagnetic wave energy for a coverage range (or coverage angle range) corresponding to a different direction (e.g., a second direction ({circle around (2)}) or a third direction ({circle around (3)})) or may expand the effective area in which electromagnetic waves can be transmitted and/or received, compared to a comparative example in which the first dielectric resonator is omitted.

[0206] According to an embodiment, a third radiation field (or a third electromagnetic field) may be generated from the first patch resonator of the first radiating portion **710** when the communication circuit **54** has a third feeding that provides an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the third feeding portion **FP3**.

[0207] According to an embodiment, the third radiation field generated during the third feeding to the third feeding portion **FP3** may provide a fourth beam pattern **714** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a first direction {circle around (1)} substantially directed toward the first surface **511**.

[0208] According to an embodiment, the third radiation field generated during the third feeding to the third feeding portion **FP3** and the fourth beam pattern **714** provided by the third radiation field, for example, may correspond to a second linear polarization having a substantially second polarization direction (e.g., a substantially seventh direction **502**).

[0209] According to an embodiment, when the communication circuit **54** has a first feeding providing an electromagnetic signal (or a radio signal, RF signal, or radiated current) to a first feeding portion **FP1**, and a third feeding providing an electromagnetic signal to a third feeding portion **FP2**, a first radiation field (or a first electromagnetic field) may be generated from a first dielectric resonator of the first radiating portion **710**, and a third radiation field (or a third electromagnetic field) may be generated from a first patch resonator of the first radiating portion **710**. Because of interference (e.g., constructive interference and/or destructive interference) between the first radiation field and the third radiation field, a fifth beam pattern **715** or a sixth beam pattern **716** may be provided, which is capable of radiating or concentrating a relatively more electromagnetic wave energy (or wave energy) in a direction different from the first direction {circle around (1)} (or tilted with respect to the first direction {circle around (1)}).

[0210] According to an embodiment, the communication circuit **54** may adjust the phase difference of the electromagnetic signal provided to the first feeding portion **FP1** and the electromagnetic signal provided to the third feeding portion **FP3**. The interference between the first radiation field generated from the first dielectric resonator of the first radiating portion **710** and the third radiation field generated from the first patch resonator of the first radiating portion **710** may provide a fifth beam pattern **715** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a fourth direction different from the first direction {circle around (1)} because of the third phase difference between the electromagnetic signals provided to the first feeding portion **FP1** and the electromagnetic signals provided to the third feeding portion **FP3**. The interference between the first radiation field generated from the first dielectric resonator of the first radiating portion **710** and the third radiation field generated from the first patch resonator of the first radiating portion **710** may provide a sixth beam pattern **716** that can radiate or concentrate a relatively more electromagnetic wave energy (or wave energy) in a fifth direction different from the first direction {circle around (1)} and the fourth direction because of the phase difference between the electromagnetic signals provided to the first feeding portion **FP1** and the electromagnetic signals provided to the third feeding portion **FP3**.

[0211] According to an embodiment, the fourth direction corresponding to the fifth beam pattern **715** and the fifth direction corresponding to the sixth beam pattern **716** may be substantially perpendicular to the seventh direction **502** (e.g., the y-axis direction) and may form an acute or obtuse angle with the first direction {circle around (1)} (e.g., the -z-axis direction). The fourth

direction and the fifth direction may be provided substantially symmetrically when viewed from the sixth direction **501**.

[0212] According to an embodiment, the angle formed by the fourth direction or the fifth direction with respect to the first direction {circle around (1)} may be substantially equal to the angle formed by the second direction {circle around (2)} or the third direction {circle around (3)} with respect to the first direction {circle around (1)}.

[0213] According to an embodiment, for the fifth beam pattern **715**, the phase difference between the electromagnetic signal provided to the first feeding portion **FP1** and the electromagnetic signal provided to the third feeding portion **FP3** may be substantially -90° . For the sixth beam pattern **716**, the phase difference between the electromagnetic signal provided to the first feeding portion **FP1** and the electromagnetic signal provided to the third feeding portion **FP3** may be substantially $+90^\circ$.

[0214] According to an embodiment, the fifth beam pattern **715** or sixth beam pattern **716** radiated during the first feeding to the first feeding portion **FP1** and the third feeding to the third feeding portion **FP3** may correspond to a second linear polarization having a substantially second polarization direction (e.g., a substantially seventh direction **502**). The first radiating portion **710** including the first dielectric resonator and the first patch resonator may extend coverage (or beam coverage), compared to a comparative example in which the first dielectric resonator is omitted. The first radiating portion **710** including the first dielectric resonator and the first patch resonator may concentrate electromagnetic wave energy for a coverage range (or coverage angle range) corresponding to a different direction (e.g., a fourth direction or a fifth direction) or may expand the effective area in which electromagnetic waves can be transmitted and/or received, compared to a comparative example in which the first dielectric resonator is omitted.

[0215] According to an embodiment, in substantially the same manner as the first radiating portion **710** provided by the combination of the first dielectric resonator including the first dielectric **61** and the first patch resonator including the first antenna element **521**, the antenna module **5** may provide the second radiating portion (or second resonator or second resonant portion) **720** provided by the combination of the second dielectric resonator (or second dielectric resonant portion) including the second dielectric **62** and the resonator (hereinafter, “second patch resonator” or “second patch resonant portion”) including the second antenna element **522**. The first beam pattern **711**, the second beam pattern **712**, the third beam pattern **713**, the fourth beam pattern **714**, the fifth beam pattern **715**, and/or the sixth beam pattern **716** may be generated from the first radiator **710** in a manner substantially identical or similar to the way in which the first beam pattern **711**, the second beam pattern **712**, the third beam pattern **713**, the fourth beam pattern **714**, the fifth beam pattern **715**, and/or the sixth beam pattern **716** are generated from the first radiating portion **710**.

[0216] According to an embodiment, in substantially the same manner as the first radiating portion **710** provided by the combination of the first dielectric resonator including the first dielectric **61** and the first patch resonator including the first antenna element **521**, the antenna module **5** may provide a third radiating portion (or third resonator or third resonant portion) **730** provided by the combination of the third dielectric resonator (or third dielectric resonant portion) including the third dielectric **63** and the resonator (hereinafter, “third patch resonator” or “third patch resonant portion”) including the third antenna element **523**. The first beam pattern **731**, the second beam pattern **732**, the third beam pattern **733**, the fourth beam pattern **734**, the fifth beam pattern **735**, and/or the sixth beam pattern **736** may be generated from the third radiator **730** in a manner substantially identical to or similar to the way in which the first beam pattern **711**, the second beam pattern **712**, the third beam pattern **713**, the fourth beam pattern **714**, the fifth beam pattern **715**, and/or the sixth beam pattern **716** are generated from the first radiating portion **710**.

[0217] According to an embodiment, in substantially the same manner as the first radiating portion **710** provided by the combination of the first dielectric resonator including the first dielectric **61** and the first patch resonator including the first antenna element **521**, the antenna module **5** may provide

a fourth radiating portion (or fourth resonator or fourth resonant portion) **740** provided by the combination of the fourth dielectric resonator (or fourth dielectric resonant portion) including the fourth dielectric **64** and the resonator (hereinafter, “fourth patch resonator” or “fourth patch resonant portion”) including the fourth antenna element **524**. The first beam pattern **741**, the second beam pattern **742**, the third beam pattern **743**, the fourth beam pattern **744**, the fifth beam pattern **745**, and/or the sixth beam pattern **746** may be generated from the fourth radiator **740** in a manner substantially identical or similar to the way in which the first beam pattern **711**, the second beam pattern **712**, the third beam pattern **713**, the fourth beam pattern **714**, the fifth beam pattern **715**, and/or the sixth beam pattern **716** are generated from the first radiating portion **710**.

[0218] According to an embodiment, because of the interference (e.g., constructive interference and/or destructive interference) between the plurality of first beam patterns **711**, **721**, **731**, and **741** during the second feeding to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**, the transmission and/or reception performance of the first linear polarization may be secured or improved for a coverage area corresponding to the first direction {circle around (1)}.

[0219] According to an embodiment, because of the phase difference between the first feeding and the second feeding for the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**, the interference (e.g., constructive interference and/or destructive interference) between the plurality of second beam patterns **712**, **722**, **732**, and **742** may result in securing or improving the transmission and/or reception performance of the first linear polarization for a coverage area corresponding to the second direction {circle around (2)}.

[0220] According to an embodiment, because of the phase difference between the first feeding and the second feeding for the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**, the interference between the plurality of third beam patterns **713**, **723**, **733**, and **743** (e.g., constructive interference and/or destructive interference) between the plurality of third beam patterns **713**, **723**, **733**, and **743**, may result in securing or improving the transmission and/or reception performance of the first linear polarization for a coverage area corresponding to the third direction {circle around (3)}.

[0221] According to an embodiment, the interference (e.g., constructive interference and/or destructive interference) between the plurality of fourth beam patterns **714**, **724**, **734**, and **744** during the third feeding to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740** may result in securing or improving the transmission and/or reception performance of the second linear polarization for the coverage area corresponding to the first direction **1**.

[0222] According to an embodiment, because of the phase difference between the first feeding and the third feeding for the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**, the interference (e.g., constructive interference and/or destructive interference) between the plurality of fifth beam patterns **715**, **725**, **735**, and **745**, the transmission and/or reception performance of the second linear polarization may be secured or improved for a coverage area corresponding to the fourth direction.

[0223] According to an embodiment, because of the phase difference between the first feeding and the third feeding for the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**, the interference (e.g., constructive interference and/or destructive interference) between the plurality of sixth beam patterns **716**, **726**, **736**, and **746**, the transmission and/or reception performance of the second linear polarization may be secured or improved for a coverage area corresponding to the fifth direction.

[0224] FIG. **9** are diagrams illustrating a distribution of radiation fields (or electromagnetic fields) generated from the first radiating portion **710** (see FIG. **7**) according to an embodiment of the disclosure. FIG. **10** are graphs illustrating radiation patterns of the first radiating portion **710**

according to an embodiment of the disclosure. FIG. 10 are graphs illustrating radiation patterns of the first radiating portion according to an embodiment of the disclosure.

[0225] Reference to FIGS. 5, 6, 9, and 10, when there are a first feeding providing an electromagnetic signal (or a radio signal, RF signal, or radiated current) to a first feeding portion FP1, and a second feeding providing an electromagnetic signal to a second feeding portion FP2, a first radiation field (or a first electromagnetic field) may be generated from a first dielectric resonator of the first radiating portion 710, and a second radiation field (or a second electromagnetic field) may be generated from a first patch resonator of the first radiating portion 710.

[0226] In an embodiment, the electric field included in the first radiation field generated during the first feeding may be provided in TE011 mode, and the electric field included in the second radiation field generated during the second feeding may be provided in TE111 mode.

[0227] According to an embodiment, the communication circuit 54 may adjust the phase of the first feeding and the phase of the second feeding such that the phase difference between the first feeding and the second feeding to the first radiating portion 710 is substantially -90° . '9A' is a diagram illustrating a distribution of an electric field in a first radiation field. '9B' is a diagram illustrating a distribution of a magnetic field in the first radiation field. '9C' is a diagram illustrating the distribution of an electric field in a second radiation field. '9D' is a diagram illustrating the distribution of a magnetic field in the second radiation field. Because of the interference (e.g., constructive interference and/or destructive interference) between the first radiation field and the second radiation field, a fourth radiation field may be provided. '9F' is a diagram illustrating a distribution of an electric field in the fourth radiation field. '9G' is a diagram illustrating the distribution of a magnetic field in the fourth radiation field. '9H' is a diagram illustrating the distribution of the wave energy in the fourth radiation field. Since the phase difference between the TE011 mode and the TE111 mode is substantially -90° , a relatively more wave energy may be radiated or concentrated for the coverage area corresponding to the second direction {circle around (2)}. '1002' of FIG. 10 is a graph illustrating a second radiation pattern (e.g., the second beam pattern 712 of FIG. 8) for a coverage area corresponding to the second direction {circle around (2)}.

[0228] According to an embodiment, the communication circuit 54 may adjust the phase of the first feeding and the phase of the second feeding such that the phase difference between the first feeding and the second feeding for the first radiating portion 710 is substantially $+90^\circ$. '9I' is a diagram illustrating a distribution of the wave energy of the fourth radiation field when the phase difference between the first feeding and the second feeding is substantially $+90^\circ$. Since the phase difference between the TE011 mode and the TE111 mode is substantially $+90^\circ$, a relatively more wave energy may be radiated or concentrated for the coverage area corresponding to the third direction {circle around (3)}. '1003' of FIG. 10 is a graph illustrating a third radiation pattern (e.g., the third beam pattern 713 of FIG. 8) for a coverage area corresponding to the third direction {circle around (3)}.

[0229] According to an embodiment, '9J' is a diagram illustrating a distribution of the wave energy of a second radiation field (or a second electromagnetic field) generated from a first patch resonator of the first radiation portion 710 when there is a second feeding to the first radiation portion 710. The wave energy may be relatively more radiated or concentrated for a coverage area corresponding to the first direction {circle around (1)}. '1001' of FIG. 10 is a graph illustrating a first radiation pattern (e.g., the first beam pattern 711 of FIG. 8) for a coverage area corresponding to the first direction {circle around (1)}.

[0230] According to an embodiment, the first radiation pattern 1001 generated during the second feeding, and the second radiation pattern 1002 or third radiation pattern 1003 generated during the first feeding and the second feeding, may substantially correspond to a first linear polarization of the first polarization direction.

[0231] According to an embodiment, in substantially the same or similar manner as the first

radiation pattern **1001** (e.g., the first beam pattern **711** of FIG. **8**) is generated during the second feeding to the second feeding portion **FP2**, a fourth radiation pattern (e.g., the fourth beam pattern **714** of FIG. **8**) may be generated during the third feeding to the third feeding portion **FP3**. In substantially the same or similar manner as the second radiation pattern **1002** (e.g., the second beam pattern **712** of FIG. **8**) or the third radiation pattern **1003** (e.g., the third beam pattern **713** of FIG. **8**) are generated during the first feeding to the first feeding portion **FP1** and the second feeding to the second feeding portion **FP2**, the fifth radiation pattern (e.g., the fifth beam pattern **715** of FIG. **8**) or the sixth radiation pattern (e.g., the sixth beam pattern **716** of FIG. **8**) may be generated during the first feeding to the first feeding portion **FP1** and the third feeding to the third feeding portion **FP3**. The fourth radiation pattern generated during the third feeding and the fifth radiation pattern or the sixth radiation pattern generated during the first feeding and the third feeding may substantially correspond to a second linear polarization of the second polarization direction.

[0232] According to an embodiment, the first radiating portion **710** may further provide a second radiation pattern **1002** (e.g., the second beam pattern **712** of FIG. **8**), a third radiation pattern **1003** (e.g., the third beam pattern **713** of FIG. **8**), a fifth radiation pattern (e.g., the fifth beam pattern **715** of FIG. **8**), or a sixth radiation pattern (e.g., the sixth beam pattern **716** of FIG. **8**), thereby extending the coverage of the antenna module **5**. The remaining radiating portions of the antenna module **5** (e.g., the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740** of FIG. **7**) may radiate a plurality of radiation patterns in substantially the same or similar manner as the first radiating portion **710**.

[0233] FIG. **11** are heat maps illustrating wave energy according to a feeding to an antenna module **5** in free space according to an embodiment of the disclosure.

[0234] Reference to FIGS. **7** and **11**, '1111' is a heat map illustrating the wave energy in a first polarization direction radiated during a second feeding to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the second feeding, for example, a plurality of first beam patterns **711**, **721**, **731**, and **741** shown in FIG. **8** may be generated. '1112' is a heat map illustrating the wave energy in the first polarization direction radiated during the first feeding and the second feeding at a -90° phase difference with respect to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the first feeding and the second feeding with a -90° phase difference, for example, the plurality of second beam patterns **712**, **722**, **732**, and **742** shown in FIG. **8** may be generated. '1113' is a heat map illustrating the wave energy in the first polarization direction radiated during the first feeding and the second feeding with a $+90^\circ$ phase difference with respect to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the first feeding and the second feeding with a $+90^\circ$ phase difference, for example, the plurality of third beam patterns **713**, **723**, **733**, and **743** shown in FIG. **8** may be generated. '1114' is a heat map illustrating the composite of the wave energy in the first polarization direction radiated during the second feeding, the wave energy in the first polarization direction radiated during the first feeding and the second feeding with a -90° phase difference, and the wave energy in the first polarization direction radiated during the first feeding and the second feeding with a $+90^\circ$ phase difference.

[0235] Reference to FIGS. **7** and **11**, '1121' is a heat map illustrating the wave energy in the second polarization direction radiated during a third feeding with respect to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the third feeding, for example, a plurality of fourth beam patterns **714**, **724**, **734**, and **744** shown in FIG. **8** may be generated. '1122' is a heat map illustrating the wave energy in the second polarization direction radiated during the first feeding and the third feeding with a -90° phase difference with respect to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the first feeding and the

third feeding with a -90° phase difference, for example, a plurality of fifth beam patterns **715**, **725**, **735**, and **745** shown in FIG. **8** may be generated. '1123' is a heat map illustrating the wave energy in the second polarization direction radiated during the first feeding and the third feeding with a $+90^\circ$ phase difference with respect to the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740**. During the first feeding and the third feeding with a $+90^\circ$ phase difference, for example, the plurality of sixth beam patterns **716**, **726**, **736**, and **746** shown in FIG. **8** may be generated. '1124' is a heat map illustrating the composite of the wave energy in the second polarization direction radiated during the third feeding, the wave energy in the second polarization direction radiated during the first feeding and the third feeding with a -90° phase difference, and the wave energy in the second polarization direction radiated during the first feeding and the third feeding with a $+90^\circ$ phase difference.

[0236] Table 1 illustrates the gain of the antenna module **5** for electromagnetic waves in the first polarization direction and electromagnetic waves in the second polarization direction, at an operating frequency of about 28 GHz.

TABLE-US-00001 TABLE 1 Electromagnetic waves of the first of the second polarization direction polarization direction Beam during Total Beam during Total 28 GHz second feeding beams second feeding beams Peak 11.7 11.7 11.0 11.0 Cumulative 1.0 4.9 1.5 4.7 distribution function (CDF) 50% Unit: dB

[0237] Referring to FIGS. **7** and **11** and Table 1, the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and the fourth radiating portion **740** of the antenna module **5** further utilize dielectric resonators in the patch resonator to extend coverage. For example, on the basis of CDF 50%, a radiating portion provided by a combination of a dielectric resonator and a patch resonator may have an improved performance of about 3.9 dB for an electromagnetic wave in the first polarization direction, compared to a patch resonator. For example, on the basis of CDF 50%, a radiating portion provided by a combination of a dielectric resonator and a patch resonator may have an improved performance of about 3.2 dB for an electromagnetic wave in the second polarization direction, compared to a patch resonator.

[0238] Table 2 illustrates the gain for the antenna module of the comparative example and the gain for the antenna module **5** of the disclosure (see FIG. **5**) in free space.

TABLE-US-00002 TABLE 2 Division 25 GHz 28 GHz 38.5 GHz CDF CDF CDF 50% Peak 50% Peak 50% Peak Antenna module for First linear 3.3 9.2 2.6 11.8 1.4 13.3 comparison example polarization Second linear 2.8 9.9 2.1 11.1 0.5 12.2 polarization Antenna module 5 of First linear 6.1 10.1 5.0 11.8 3.8 13.3 the disclosure polarization Second linear 6.5 11.5 5.0 11.1 4.1 12.6 polarization Unit: dB

[0239] Table 3 illustrates the gain for the antenna module of the comparative example in an electronic device (e.g., the foldable electronic device **3** of FIG. **3**) and the gain for the antenna module **5** of the disclosure (see FIG. **5**).

TABLE-US-00003 TABLE 3 Division 25 GHz 28 GHz 38.5 GHz CDF CDF CDF 50% Peak 50% Peak 50% Peak Antenna module 1st linear -0.7 10.2 -1.0 11.4 -1.9 10.3 for comparison polarization example Second linear -0.4 8.7 -1.2 11.0 -2.4 9.9 polarization Antenna module 1st linear 1.6 12.5 0.8 11.4 0.2 10.4 5 of the polarization disclosure Second linear 2.0 11.6 1.4 11.0 -0.4 10.5 polarization Unit: dB

[0240] The antenna module **5** according to an embodiment of the disclosure may include a plurality of radiating portions (e.g., the first radiating portion **710**, the second radiating portion **720**, the third radiating portion **730**, and/or the fourth radiating portion **740** of FIG. **7**) provided by a combination of dielectric resonators and patch resonators. The antenna module of the comparative example may be provided with the dielectric resonator omitted, compared to the antenna module **5** according to an embodiment of the disclosure.

[0241] With reference to Tables 2 and 3, the antenna module **5** according to an embodiment of the disclosure may provide superior coverage compared to the antenna module of the comparative

example. For example, in free space, the antenna module **5** may have an improved performance of about 2.4 to about 3.7 dB, on the basis of a CDF of 50%, compared to the antenna module of the comparative example. For example, in an electronic device (e.g., the foldable electronic device **3** of FIG. **3**), the antenna module **5** may have an improved performance of about 1.8 to about 2.6 dB, on the basis of a CDF of 50%, compared to the antenna module of the comparative example.

[0242] FIG. **12** is an exploded perspective view of an antenna module **12** according to an embodiment of the disclosure. FIG. **13** is an exploded perspective view of the antenna module **12** according to an embodiment of the disclosure.

[0243] Referring to FIGS. **12** and **13**, the antenna module **12** may include a printed circuit board (or antenna structure) **1210** (e.g., the printed circuit board **510** of FIG. **5**), a first dielectric **1221** (e.g., the first dielectric **61** of FIG. **5**), a second dielectric **1222** (e.g., the second dielectric **62** of FIG. **5**), a third dielectric **1223** (e.g., the third dielectric **63** of FIG. **5**), a fourth dielectric **1224** (e.g., the fourth dielectric **64** of FIG. **5**), a first connector **1231**, a second connector **1232**, a third connector **1233**, a fourth connector **1234**, and/or a communication circuit (e.g., the communication circuit **54** of FIG. **6**).

[0244] According to an embodiment, the printed circuit board **1210** may include a first surface **1210a** (e.g., the first surface **511** of FIG. **5**) and a second surface (e.g., the second surface **512** of FIG. **5**) facing in an opposite direction from the first surface **1210a**. The printed circuit board **1210** may include a first antenna element (or a first patch) **1211** (the first antenna element **521** of FIG. **5**), a second antenna element (or a second patch) **1212** (the second antenna element **522** of FIG. **5**), a third antenna element (or third patch) **1213** (the third antenna element **523** of FIG. **5**), and/or a fourth antenna element (or fourth patch) **1214** (the fourth antenna element **524** of FIG. **5**). The antenna array (e.g., the antenna array **52** of FIG. **5**) including the first antenna element **1211**, the second antenna element **1212**, the third antenna element **1213**, and the fourth antenna element **1214** may be disposed on the first surface **1210a**, or may be disposed on the interior of the printed circuit board **1210** closer to the first surface **1210a** than the second surface.

[0245] According to an embodiment, the communication circuit (e.g., the communication circuit **54** of FIG. **6**) may be disposed on the second surface (e.g., the second surface **512** of FIG. **6**) of the printed circuit board **1210**.

[0246] According to an embodiment, the first dielectric **1221**, the second dielectric **1222**, the third dielectric **1223**, and the fourth dielectric **1224** may be disposed on or coupled to the first surface **1210a** of the printed circuit board **1210**. The first dielectric **1221** may be positioned corresponding to the first antenna element **1211**. When viewed from above the first surface **1210a** (e.g., when viewed in the +z axis direction), the first dielectric **1221** may overlap with the first antenna element **1211**. The second dielectric **1222** may be positioned corresponding to the second antenna element **1212**. When viewed from above the first surface **1210a**, the second dielectric **1222** may be overlapped with the second antenna element **1212**. The third dielectric **1223** may be positioned corresponding to the third antenna element **1213**. When viewed from above the first surface **1210a**, the third dielectric **1223** may overlap with the third antenna element **1213**. The fourth dielectric **1224** may be positioned corresponding to the fourth antenna element **1214**. When viewed from above the first surface **1210a**, the fourth dielectric **1224** may overlap with the fourth antenna element **1214**.

[0247] According to an embodiment, a first connector **1231** may be positioned corresponding to the first antenna element **1211** and the first dielectric **1221**, and may be disposed at least partially between the printed circuit board **1210** and the first dielectric **1221**. A second connector **1232** may be positioned corresponding to the second antenna element **1212** and the second dielectric **1222**, and may be disposed at least partially between the printed circuit board **1210** and the second dielectric **1222**. A third connector **1233** may be positioned corresponding to the third antenna element **1213** and the third dielectric **1223**, and may be disposed at least partially between the printed circuit board **1210** and the third dielectric **1223**. A fourth connector **1234** may be positioned

corresponding to the fourth antenna element **1214** and the fourth dielectric **1224**, and may be disposed at least partially between the printed circuit board **1210** and the fourth dielectric **1224**. [0248] According to an embodiment, the first dielectric **1221** may include a first portion **1221a** and a plurality of second portions **1221b**, **1221c**, **1221d**, and **1221e**. The first portion **1221a** may overlap with the first antenna element **1211** when viewed from above the first surface **1210a**. The first connector **1231** may be positioned between the first portion **1221a** and the printed circuit board **1210**. The plurality of second portions **1221b**, **1221c**, **1221d**, and **1221e** may extend from the first portion **1221a** and may be in contact with or be coupled to the first surface **1210a** of the printed circuit board **1210**. The plurality of second portions **1221b**, **1221c**, **1221d**, and **1221e** (e.g., ribs or bridges) may support spacing between the first portion **1221a** and the first surface **1210a** of the printed circuit board **1210**. The second dielectric **1222**, third dielectric **1223**, or fourth dielectric **1224** may be provided substantially the same as or similar to the first dielectric **1221**.

[0249] According to an embodiment, the first antenna element **1211** may include a first opening **1211a** (e.g., the opening **5211** of FIG. 5). The second antenna element **1212** may include a second opening **1212a**. The third antenna element **1213** may include a third opening **1213a**. The fourth antenna element **1214** may include a fourth opening **1214a**.

[0250] According to an embodiment, the printed circuit board **1210** may include a first conductive via **V1** positioned corresponding to a first opening **1211a** of the first antenna element **1211**, a second conductive via **V2** positioned corresponding to a second opening **1212a** of the second antenna element **1212**, a third conductive via **V3** positioned corresponding to the third opening **1213a** of the third antenna element **1213**, and/or a fourth conductive via **V4** positioned corresponding to a fourth opening **1214a** of the fourth antenna element **1214**. The first conductive via **V1** may penetrate the first opening **1211a**. The second conductive via **V2** may penetrate the second opening **1212a**. A third conductive via **V3** may penetrate the third opening **1213a**. The fourth conductive via **V4** may penetrate the fourth opening **1214a**. The first conductive via **V1** may be part of a first electrical path (or first conductive path) connecting the first dielectric **1221** and a communication circuit (e.g., the communication circuit **54** of FIG. 6). The second conductive via **V2** may be part of a second electrical path (or second conductive path) connecting the second dielectric **1222** and the communication circuit (e.g., the communication circuit **54** of FIG. 6). The third conductive via **V3** may be part of a third electrical path (or third conductive path) connecting the third dielectric **1223** and the communication circuit (e.g., the communication circuit **54** of FIG. 6). The fourth conductive via **V4** may be part of a fourth electrical path (or fourth conductive path) connecting the fourth dielectric **1224** and the communication circuit (e.g., the communication circuit **54** of FIG. 6).

[0251] According to an embodiment, the first connector **1231** may include a first conductive portion **C1** connecting the first conductive via **V1** and the first dielectric **1221**. One end of the first conductive portion **C1** may be electrically connected to the first conductive via **V1** via a conductive adhesive material or conductive bonding material such as solder. The first conductive portion **C1** may be part of a first electrical path (or first conductive path) connecting the first dielectric **1221** and a communication circuit (e.g., the communication circuit **54** of FIG. 6).

[0252] According to an embodiment, the other end of the first conductive portion **C1** may be in physical contact or direct connection with the first dielectric **1221**. During the first feeding, the first dielectric **1221** may receive an electromagnetic signal directly from the first conductive portion **C1**.

[0253] According to an embodiment, the first connector **1231** may include a first non-conductive portion **12311** including a non-conductive material, such as a polymer. The first conductive portion **C1** may be disposed on the first non-conductive portion **12311**.

[0254] According to an embodiment, the first connector **1231** may be implemented as a printed circuit board. For example, the first non-conductive portion **12311** may be a non-conductive layer (e.g., a dielectric layer) of a printed circuit board, and the first conductive portion **C1** may be a conductive via provided on the non-conductive layer.

[0255] According to various embodiments, the first conductive portion C1 may be connected to the first dielectric **1221** indirectly through the first non-conductive portion **12311**. The first dielectric **1221** and the first non-conductive portion **12311** may be in physical contact. During the first feeding by a communication circuit (e.g., the communication circuit **54** of FIG. **6**), the first dielectric **1221** may indirectly receive an electromagnetic signal from the first conductive portion C1 through the first non-conductive portion **12311**. In various embodiments, the combination of the first dielectric **1221** and the first non-conductive portion **12311** may be interpreted as the first dielectric **61** of FIG. **5**.

[0256] According to various embodiments, the antenna module **12** may include a non-conductive adhesive material or a non-conductive bonding material disposed between the first dielectric **1221** and the first non-conductive portion **12311**. The first dielectric **1221** and the first non-conductive portion **12311** may be coupled through the non-conductive adhesive material or a non-conductive bonding material. During the first feeding, the first dielectric **1221** may indirectly receive electromagnetic signals from the first conductive portion C1 through the first non-conductive portion **12311** and the non-conductive adhesive material (or the non-conductive bonding material). In various embodiments, the combination of the first dielectric **1221**, the first non-conductive portion **12311**, and the non-conductive adhesive material (or the non-conductive bonding material) may be interpreted as the first dielectric **61** of FIG. **5**.

[0257] According to an embodiment, the second connector **1232**, third connector **1233**, or fourth connector **1234** may be provided substantially the same as or similar to the first connector **1231** connecting the printed circuit board **1210** and the first dielectric **1221**. The second connector **1232** may include a second non-conductive portion **12321** and a second conductive portion C2 disposed on the second non-conductive portion **12321**. The second conductive portion C2 may be electrically connected to the second conductive via V2 of the printed circuit board **1210** through a conductive adhesive material or a conductive bonding material such as solder. The third connector **1233** may include a third non-conductive portion **12331** and a third conductive portion C3 disposed on the third non-conductive portion **12331**. The third conductive portion C3 may be electrically connected to the third conductive via V3 of the printed circuit board **1210** through a conductive adhesive material or a conductive bonding material such as solder. The fourth connector **1234** may include a fourth non-conductive portion **12341** and a fourth conductive portion C4 disposed on the fourth non-conductive portion **12341**. The fourth conductive portion C4 may be electrically connected to the fourth conductive via V4 of the printed circuit board **1210** through a conductive adhesive material or a conductive bonding material such as solder.

[0258] FIG. **14** is a cross-sectional view of antenna module **12** as viewed along line A-A' in FIG. **13** according to an embodiment of the disclosure.

[0259] Referring to FIG. **14**, the antenna module **12** may include a printed circuit board **1210**, a first dielectric **1221**, and/or a first connector **1231**.

[0260] According to an embodiment, the printed circuit board **1210** may include a first antenna element (or a first patch) **1211**, a ground plane **1216** (e.g., the ground plane **53** of FIG. **5**), a dielectric layer (e.g., an insulating layer) **1215**, and/or a first conductive via V1. The first antenna element **1211** and the ground plane **1216** may be disposed in the dielectric layer **1215**. The first antenna element **1211** may be positioned closer to the first surface **1210a** than the ground plane **1216**. In an embodiment, the first antenna element **1211** and the ground plane **1216** may be positioned physically spaced apart from each other. When viewed from above a first surface **1210a** of a printed circuit board **1210**, a first antenna element **1211** may overlap with a ground plane **1216**. The first antenna element **1211** may include a first opening **1211a**. When viewed from above the first surface **1210a** of the printed circuit board **1210**, a first conductive via V1 may overlap with the first opening **1211a**. The first conductive via V1 may be a part of a signal line (not shown separately) electrically connected to a communication circuit (e.g., the communication circuit **54** of FIG. **6**) of the printed circuit board **1210**. The signal line may be positioned to be physically spaced

from the ground plane **1216**.

[0261] According to an embodiment, the first connector **1231** may include a first non-conductive portion **12311** and a first conductive portion **C1**. **1** The first connector **1231** may be provided, for example, as a printed circuit board. The first non-conductive portion **12311** may be a non-conductive layer (e.g., a dielectric layer) of the printed circuit board, and the first conductive portion **C1** may be a conductive via provided in the non-conductive layer.

[0262] According to an embodiment, the first connector **1231** may be coupled to a printed circuit board **1210**. **1** The first conductive portion **C1** of the connector **1231** may be electrically connected to the first conductive via **V1** of the printed circuit board **1210** through a conductive adhesive material or a conductive bonding material such as solder.

[0263] According to an embodiment, the first portion **1221a** of the first dielectric **1221** may be in physical contact with the first conductive portion **C1**.

[0264] According to various embodiments, an integral printed circuit board may be provided to replace the printed circuit board **1210** and the first connector **1231**.

[0265] According to various embodiments, a metal portion (or a metal body) such as a metal pin (not shown) may be provided to replace the first connector **1231**. For example, the metal portion may be connected to the first conductive via **V1** through a conductive adhesive material or a conductive bonding material such as solder. The metal portion may be in physical contact with the first dielectric **1221**. In various embodiments, the metal portion may be inserted into a hole or recess in the first dielectric **1221** (not shown).

[0266] In various embodiments, an integral dielectric (not shown separately) may be provided to replace at least two of the first dielectric **1221**, the second dielectric **1222**, the third dielectric **1223**, and the fourth dielectric **1224** of FIG. **13**.

[0267] According to various embodiments, the first dielectric **1221**, the second dielectric **1222**, the third dielectric **1223**, and the fourth dielectric **1224** of FIG. **13** may be a plurality of non-conductive material members (not shown separately) disposed on the first cover **312** (see FIG. **3**) of the foldable electronic device **3**.

[0268] According to various embodiments, the first cover **312** (see FIG. **3**) of the foldable electronic device **3** may replace the first dielectric **1221**, the second dielectric **1222**, the third dielectric **1223**, and the fourth dielectric **1224** of FIG. **13**.

[0269] FIG. **15** is a cross-sectional view of an antenna module of another embodiment that is a variation of the embodiment of FIG. **14** according to an embodiment of the disclosure.

[0270] Referring to FIG. **15**, the antenna module **15** may include a printed circuit board **1510** and a first dielectric **1521**.

[0271] According to an embodiment, the printed circuit board **1510** may include a first antenna element (or a first patch) **1511** (e.g., the first antenna element **521** of FIG. **5**), a ground plane **1516** (e.g., the ground plane **53** of FIG. **5**), a dielectric layer (e.g., an insulating layer) **1515**, and/or a first conductive via **V11**. The first antenna element **1511** and the ground plane **1516** may be disposed on the dielectric layer **1515**. The first antenna element **1511** may be positioned closer to the first surface **1510a** (e.g., the first surface **511** of FIG. **5**) than the ground plane **1516**. In an embodiment, the first antenna element **1511** and the ground plane **1516** may be physically spaced apart. When viewed from above the first surface **1510a**, the first antenna element **1511** may overlap with the ground plane **1516**. The first antenna element **1511** may include a first opening **1511a**. When viewed from above the first surface **1510a**, the first conductive via **V11** may overlap with the first opening **1511a**. The first conductive via **V11** may be part of a signal line (not shown separately) electrically connected to a communication circuit (e.g., the communication circuit **54** of FIG. **6**) of the printed circuit board **1510**. The signal lines may be disposed physically spaced apart from the ground plane **1516**.

[0272] According to an embodiment, the first dielectric **1521** may be disposed on the first surface **1410a** of the printed circuit board **1510**. The first dielectric **1521** may be in physical contact with

the first conductive via **V11**.

[0273] According to an embodiment, the first antenna element **1511** of the printed circuit board **1510** may be disposed on an interior of the printed circuit board **1510** such that it is spaced apart from the first dielectric **1521** disposed on the first surface **1510a**.

[0274] According to various embodiments, the antenna module **15** may further include a metal portion (or a metal body) (e.g., a metal pin, not shown separately) directly coupled to the first conductive via **V11**. For example, the metal portion may be coupled to the first conductive via **V11** through a conductive adhesive material or a conductive bonding material such as solder. The metal portion may be in physical contact with the first dielectric **1521**. In various embodiments, the metal portion may be inserted into a recess of the first dielectric **1521** (not shown).

[0275] In various embodiments, the first cover **312** of the foldable electronic device **3** (see FIG. 3), or at least one member of a non-conductive material disposed on the first cover **312** (not shown separately), may replace the first dielectric **1521**.

[0276] FIG. 16 are graphs illustrating radiation patterns of a first radiating portion **1600** (e.g., the first radiating portion **710** of FIG. 7) according to the permittivity of the first dielectric **1221** in the embodiment of FIG. 13 according to an embodiment of the disclosure. FIG. 17 are graphs illustrating radiation patterns of a first radiating portion **1600** (e.g., the first radiating portion **710** of FIG. 7) along the length of the first conductive portion **C1** extending in a first direction toward the first surface **1210a** in the embodiment of FIG. 14 according to an embodiment of the disclosure.

[0277] Referring to FIGS. 13 and 16, '1601' are graphs illustrating radiation patterns of the first radiating portion **1600** when the relative permittivity of the first dielectric **1221** is substantially 1. '1602' are graphs illustrating radiation patterns of the first radiating portion **1600** when the relative permittivity of the first dielectric **1221** is substantially 3. '1603' is a graph illustrating the radiation patterns of the first radiating portion **1600** when the relative permittivity of the first dielectric **1221** is substantially 5. The length of the first dielectric **1221** extending in the first direction {circle around (1)} may be, for example, about 0.5 mm.

[0278] Referring to FIGS. 14 and 17, '1701' are graphs illustrating radiation patterns of the first radiating portion **1600** when the length of the first conductive portion **C1** extending in the first direction {circle around (1)} is about 0.5 mm. '1702' are graphs illustrating radiation patterns of the first radiating portion **1600** when the first conductive portion **C1** has a length of about 1 mm extending in the first direction {circle around (1)}. The relative permittivity of the first dielectric **1221** may be substantially 1. For example, it can be interpreted that as the length of the first conductive portion **C1** extending in the first direction {circle around (1)} increases, the length at which the feeding path is inserted into the first dielectric **1221** increases.

[0279] According to an embodiment, during the second feeding to the first radiating portion **1600**, relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the first direction {circle around (1)}. '1611', '1621', and '1631' of FIG. 16 are graphs illustrating a first radiation pattern (e.g., the first beam pattern **711** of FIG. 8) for a coverage area corresponding to the first direction {circle around (1)} during the second feeding. '1711' and '1721' of FIG. 17 are graphs illustrating a first radiation pattern (e.g., the first beam pattern **711** of FIG. 8) for a coverage area corresponding to the first direction {circle around (1)} during the second feeding.

[0280] According to an embodiment, during a first feeding and a second feeding with a -90° phase difference with respect to the first radiating portion **1600**, relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the second direction {circle around (2)} tilted with respect to the first direction {circle around (1)}. '1612', '1622', '1632' of FIG. 16 are graphs illustrating a second radiation pattern (e.g., the second beam pattern **712** of FIG. 8) for a coverage area corresponding to the second direction {circle around (2)} during the first feeding and the second feeding with a -90° phase difference. '1712' and '1722' of FIG. 17 are graphs illustrating a second radiation pattern (e.g., the second beam pattern **712** of FIG. 8) for a coverage

area corresponding to the second direction (2 during the first feeding and the second feeding with a -90° phase difference.

[0281] According to an embodiment, during the first feeding and the second feeding with a $+90^\circ$ phase difference with respect to the first radiating portion **1600**, relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the third direction {circle around (3)} tilted with respect to the first direction {circle around (1)}. '1613', '1623', and '1633' of FIG. **16** are graphs illustrating a third radiation pattern (e.g., the third beam pattern **713** of FIG. **8**) for a coverage area corresponding to the third direction 3 during the first feeding and the second feeding with a $+90^\circ$ phase difference. '1713' and '1723' of FIG. **17** are graphs illustrating a third radiation pattern (e.g., the third beam pattern **713** of FIG. **8**) for a coverage area corresponding to the third direction {circle around (3)} during first feeding and second feeding with a $+90^\circ$ phase difference. [0282] Comparing the graphs of FIG. **16**, when the relative permittivity of the first dielectric **1221** increases, the second direction {circle around (2)} or the third direction {circle around (3)} may be further tilted corresponding to the first direction {circle around (1)} during the first feeding and the second feeding, such that the coverage may be expanded.

[0283] Comparing the graphs of FIG. **17**, when the length of the first conductive portion C1 extending in the first direction {circle around (1)} increases, the second direction {circle around (2)} or the third direction {circle around (3)} may be further tilted with respect to the first direction {circle around (1)} during the first feeding and the second feeding, such that the coverage may be expanded.

[0284] FIG. **18** is a diagram illustrating an antenna module **18** according to an embodiment of the disclosure. FIG. **19** is a cross-sectional view of antenna module **18** as viewed along line B-B' in FIG. **18** according to an embodiment of the disclosure.

[0285] Referring to FIGS. **18** and **19**, the antenna module **18** may include an antenna structure **181**, a communication circuit (e.g., the communication circuit **54** of FIG. **6**), a first dielectric **1851** (e.g., the first dielectric **61** of FIG. **5**), a second dielectric **1852** (e.g., the second dielectric **62** of FIG. **5**), a third dielectric **1853** (e.g., the third dielectric **63** of FIG. **5**), and/or a fourth dielectric **1854** (e.g., the fourth dielectric **64** of FIG. **5**).

[0286] According to various embodiments, the antenna module **18** may be the third antenna module **246** of FIG. **2**.

[0287] According to an embodiment, the antenna structure **181** may include a printed circuit board **1810**. The printed circuit board **1810** may include a first surface **1811** (e.g., the first surface **511** of FIG. **5**) and a second surface **1812** (e.g., the second surface **512** of FIG. **5**) facing in an opposite direction from the first surface **1811**.

[0288] According to an embodiment, the first surface **1811** of the printed circuit board **1810** may face the first cover **312** (see FIG. **3**). The second surface **1812** of the printed circuit board **1810** may face the first display area {circle around (1)} of the first display module **34**.

[0289] According to an embodiment, the printed circuit board **1810** may include a first antenna array **1820** and a second antenna array **1830**. The second antenna array **1830** may be disposed on the first surface **1811**, or may be located closer to the first surface **1811** of the printed circuit board **1810** than the first antenna array **1820**. The first antenna array **1820** may be disposed on an interior of the printed circuit board **1810** closer to the second surface **1812** of the printed circuit board **1810** than the third antenna array **1830**.

[0290] According to an embodiment, the first antenna array **1820** may include a first antenna element **1821**, a second antenna element **1822**, a third antenna element **1823**, and/or a fourth antenna element **1824**. The second antenna array **1830** may include a fifth antenna element **1831**, a sixth antenna element **1832**, a seventh antenna element **1833**, and/or an eighth antenna element **1834**. When viewed from above the first surface **1811** (e.g., when viewed in the $+z$ axis direction), the first antenna element **1821** and the fifth antenna element **1831** may overlap. When viewed from above the first surface **1811**, the second antenna element **1822** and the sixth antenna element **1832**

may be overlapped. When viewed from above the first surface **1811**, the third antenna element **1823** and the seventh antenna element **1833** may be overlapped. When viewed from above the first surface **1811**, the fourth antenna element **1824** and the eighth antenna element **1834** may be overlapped. The first antenna array **1820** or the second antenna array **1830** may be, for example, the antenna **248** of FIG. 2.

[0291] According to an embodiment, the plurality of antenna elements **1821**, **1822**, **1823**, and **1824** of the first antenna array **1820** may be substantially the same shape and may be spaced apart. The plurality of antenna elements **1821**, **1822**, **1823**, and **1824** may be disposed in a sixth direction **1801** (e.g., the y-axis direction) when viewed from above the first surface **1811** (e.g., the +z-axis direction). The sixth direction **1801** may be substantially parallel to, for example, a center line A (see FIG. 3) of the foldable electronic device **3**.

[0292] According to an embodiment, when viewed from above the first surface **1811** (e.g., when viewed in the +z axis direction), the plurality of antenna elements **1821**, **1822**, **1823**, and **1824** of the first antenna array **1820** may be substantially square. The first antenna element **1821** may include a first edge E11, a second edge E12, a third edge E13, and a fourth edge E14, when viewed from above the first surface **1811**. When viewed from above the first surface **1811**, the first edge E11 and the second edge E12 may be spaced apart from each other in the sixth direction **1801** and may be substantially parallel to each other. When viewed from above the first surface **1811**, the third edge E13 and the fourth edge E14 may be spaced apart from each other in the seventh direction **1802** and may be substantially parallel to each other. The remaining antenna elements **1822**, **1823**, and **1824** may be substantially the same shape as the first antenna element **1821**. The shape of the plurality of antenna elements **1821**, **1822**, **1823**, and **1824** is not limited thereto and may vary. The plurality of antenna elements **1821**, **1822**, **1823**, and **1824** may be provided in various polygons, such as, for example, circular, elliptical, or triangular (not shown).

[0293] According to an embodiment, the plurality of antenna elements **1831**, **1832**, **1833**, and **1834** of the second antenna array **1830** may be substantially the same shape and may be disposed at regular intervals. The plurality of antenna elements **1821**, **1822**, **1823**, and **1824** may be disposed in the sixth direction **1801** (e.g., the y-axis direction) when viewed from above the first surface **1811** (e.g., when viewed in the +z-axis direction).

[0294] According to an embodiment, when viewed from above the first surface **1811** (e.g., when viewed in the +z axis direction), the plurality of antenna elements **1831**, **1832**, **1833**, and **1834** of the second antenna array **1830** may be substantially square. The fifth antenna element **1831** may include a fifth edge E15, a sixth edge E16, a seventh edge E17, and an eighth edge E18 when viewed from above the first surface **1811**. The fifth edge E15 may be positioned corresponding to the first edge E11 included in the first antenna element **1821** of the first antenna array **1820**. The sixth edge E16 may be positioned corresponding to the second edge E12 included in the first antenna element **1821** of the first antenna array **1820**. The seventh edge E17 may be positioned corresponding to the third edge E13 included in the first antenna element **1821** of the first antenna array **1820**. The eighth edge E18 may be positioned corresponding to the fourth edge E14 included in the first antenna element **1821** of the first antenna array **1820**. The center of the first antenna element **1821** included in the first antenna array **1820** and the center of the fifth antenna element **1831** included in the second antenna array **1830** may be substantially aligned, when viewed from above the first surface **1811**. The remaining antenna elements **1832**, **1833**, and **1834** may be provided substantially the same as the fifth antenna element **1831**. The shape of the plurality of antenna elements **1831**, **1832**, **1833**, and **1834** is not limited thereto and may vary. The plurality of antenna elements **1831**, **1832**, **1833**, and **1834** may be provided in various polygons, such as, for example, circular, elliptical, or triangular (not shown).

[0295] According to an embodiment, the plurality of antenna elements **1831**, **1832**, **1833**, and **1834** of the second antenna array **1830** may be provided in a smaller size than the plurality of antenna elements **1821**, **1822**, **1823**, and **1824** of the first antenna array **1820** when viewed from above the

first surface **1811**.

[0296] According to various embodiments, the number or location of antenna elements included in the first antenna array **1820** and the second antenna array **1830** may vary, without limitation to the embodiments shown.

[0297] According to an embodiment, the printed circuit board **1810** may include a ground plane **1840** (e.g., the ground plane **53** of FIG. **5**) that operates as an antenna ground. The ground plane **1840** may contribute to ensuring antenna radiation performance (or radio wave transmission or reception performance, or communication performance) and/or ensuring coverage, with respect to the first antenna array **1820** and the second antenna array **1830**. The ground plane **53** may reduce electromagnetic interference (EMI) (or signal loss) with respect to the first antenna array **1820** and the second antenna array **1830**.

[0298] According to an embodiment, the ground plane **1840** may be disposed closer to the second surface **1812** of the printed circuit board **1810** than the first antenna array **1820**. When viewed from above the first surface **1811** (e.g., when viewed in the +z axis direction), the first antenna array **1820** and the second antenna array **1830** may overlap at least partially with the ground plane **1840**.

[0299] According to an embodiment, the antenna structure **181** may operate as a patch antenna. For example, when a communication circuit (e.g., the communication circuit **54** of FIG. **6**) provides (or feedings) an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the plurality of antenna elements **1821**, **1822**, **1823**, and **1824** of the first antenna array **1820**, because of the plurality of antenna elements (e.g., the plurality of first patches) **1821**, **1822**, **1823**, and **1824** resonating with the ground plane **1840** and acting as radiating elements at a selected or specified first frequency (or a first frequency band), the antenna structure **181** may operate as a resonator (e.g., a patch antenna). When a communication circuit (e.g., the communication circuit **54** of FIG. **6**) provides (or feedings) an electromagnetic signal to the plurality of antenna elements **1831**, **1832**, **1833**, and **1834** of the second antenna array **1830**, because of the plurality of antenna elements (e.g., the plurality of second patches) **1831**, **1832**, **1833**, and **1834** resonating with the ground plane **1840** and acting as radiating elements at a selected or specified second frequency (or a second frequency band), the antenna structure **181** may operate as a resonator (e.g., a patch antenna).

[0300] According to an embodiment, the printed circuit board **1810** may include a second electrical path (or a second transmission line or a second feeding line) (not shown) that transmits an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the first antenna element **1821** of the first antenna array **1820**. The second electrical path may electrically connect the first antenna element **1821** and a communication circuit (e.g., the communication circuit **54** of FIG. **6**). The portion of the first antenna element **1821** to which the electromagnetic signal is transmitted from the second electrical path may be defined or interpreted as the second feeding portion **FP12**. In some embodiments, the second electrical path, or the portion of the second electrical path connected to the first antenna element **1821**, may be defined or interpreted as the 'second feeding portion', and the portion of the first antenna element **1821** to which the electromagnetic signal is transmitted from the second electrical path may be defined or interpreted as the 'second feeding area' or the 'second feeding point'. The electromagnetic signal provided by the communication circuit (e.g., the communication circuit **54** of FIG. **6**) to the second feeding portion **FP12** (or the second electrical path, the second feeding area, or the second feeding point) may be defined or interpreted as a 'second feeding'.

[0301] According to an embodiment, the printed circuit board **1810** may include a third electrical path (or a third transmission line or a third feeding line) (not shown) that transmits electromagnetic signals to the first antenna element **1821** of the first antenna array **1820**. The third electrical path may electrically connect the first antenna element **1821** and a communication circuit (e.g., the communication circuit **54** of FIG. **6**). The portion of the first antenna element **1821** to which the electromagnetic signal is transmitted from the third electrical path may be defined or interpreted as the third feeding portion **FP13**. In some embodiments, the third electrical path, or the portion of the

third electrical path connected to the first antenna element **1821**, may be defined or interpreted as the 'third feeding portion' and the portion of the first antenna element **521** to which the electromagnetic signal from the third electrical path is transmitted may be defined or interpreted as the 'third feeding area' or the 'third feeding point'. The electromagnetic signal provided by the communication circuit (e.g., the communication circuit **54** of FIG. **6**) to the third feeding portion **FP13** (or the third electrical path, the third feeding area, or the third feeding point) may be defined or interpreted as a 'third feeding'.

[0302] According to an embodiment, the second feeding portion **FP12** of the first antenna element **1821** included in the first antenna array **1820** may be positioned between the center and the first edge **E11** of the first antenna element **1821** in the sixth direction **1801** when viewed from above the first surface **1811** (e.g., when viewed in the +z axis direction). The third feeding portion **FP13** of the first antenna element **1821** included in the first antenna array **1820** may be positioned between the center and the third edge **E13** of the first antenna element **1821** in the seventh direction **1802**, when viewed from above the first surface **1811**. The distance that the second feeding portion **FP12** is spaced from the center of the first antenna element **1821** in the sixth direction **1801** and the distance that the third feeding portion **FP13** is spaced from the center of the first antenna element **1821** in the seventh direction **1802** may be substantially the same. The remaining antenna elements **1822**, **1823**, and **1824** of the first antenna array **1820** may include a second feeding portion corresponding to the second feeding portion **FP12** of the first antenna element **1821**, and a third feeding portion corresponding to the third feeding portion **FP13** of the first antenna element **1821**.

[0303] According to an embodiment, the printed circuit board **1810** may include a fourth electrical path (or a fourth transmission line or a fourth feeding line) (not shown) that transmits an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the fifth antenna element **1831** of the second antenna array **1830**. The fourth electrical path may electrically connect the fifth antenna element **1831** and a communication circuit (e.g., the communication circuit **54** of FIG. **6**). The portion of the fifth antenna element **1831** to which the electromagnetic signal is transmitted from the fourth electrical path may be defined or interpreted as the fourth feeding portion **FP14**. In some embodiments, the fourth electrical path, or the portion of the fourth electrical path connected to the fifth antenna element **1831**, may be defined or interpreted as a 'fourth feeding portion', and the portion of the fifth antenna element **1831** to which the electromagnetic signal is transmitted from the fourth electrical path may be defined or interpreted as a 'fourth feeding area' or a 'fourth feeding point'. The electromagnetic signal provided by the communication circuit (e.g., the communication circuit **54** of FIG. **6**) to the fourth feeding portion **FP14** (or the fourth electrical path, the fourth feeding area, or the fourth feeding point) may be defined or interpreted as a 'fourth feeding'.

[0304] According to an embodiment, the printed circuit board **1810** may include a fifth electrical path (or a fifth transmission line or a fifth feeding line) (not shown) that transmits electromagnetic signals to the fifth antenna element **1831** of the second antenna array **1830**. The fifth electrical path may electrically connect the fifth antenna element **1831** and a communication circuit (e.g., the communication circuit **54** of FIG. **6**). The portion of the fifth antenna element **1831** to which the electromagnetic signal is transmitted from the fifth electrical path may be defined or interpreted as the fifth feeding portion **FP15**. In some embodiments, the fifth electrical path, or the portion of the fifth electrical path connected to the fifth antenna element **1831**, may be defined or interpreted as a 'fifth feeding portion' and the portion of the fifth antenna element **1831** to which the electromagnetic signal is transmitted from the fifth electrical path may be defined or interpreted as a 'fifth feeding area' or a 'fifth feeding point'. The electromagnetic signal provided by the communication circuit (e.g., the communication circuit **54** of FIG. **6**) to the fifth feeding portion **FP15** (or the fifth electrical path, the fifth feeding area, or the fifth feeding point) may be defined or interpreted as a 'fifth feeding'.

[0305] According to an embodiment, the fourth feeding portion **FP14** of the fifth antenna element

1831 included in the second antenna array **1830** may be positioned between the center and the eighth edge **E18** of the fifth antenna element **1831** in the seventh direction **1802** when viewed from above the first surface **1811** (e.g., when viewed in the +z axis direction). The fifth feeding portion **FP15** of the fifth antenna element **1831** included in the second antenna array **1830** may be positioned between the center and the sixth edge **E16** of the second antenna element **1831** in the sixth direction **1801**, when viewed from above the first surface **1811**. The distance that the fourth feeding portion **FP14** is spaced from the center of the fifth antenna element **1831** in the seventh direction **1802** and the distance that the fifth feeding portion **FP15** is spaced from the center of the fifth antenna element **1831** in the sixth direction **1801** may be substantially the same. The remaining antenna elements **1832**, **1833**, and **1834** of the second antenna array **1830** may include a fourth feeding portion corresponding to the fourth feeding portion **FP14** of the fifth antenna element **1831**, and a fifth feeding portion corresponding to the fifth feeding portion **FP15** of the fifth antenna element **1831**.

[0306] According to an embodiment, the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and the fourth dielectric **1854** may be disposed on or coupled to the first surface **1811** of the printed circuit board **1810**.

[0307] According to an embodiment, the first dielectric **1851** may be positioned corresponding to the first antenna element **1821** of the first antenna array **1820** and the fifth antenna element **1831** of the second antenna array **1830**. The first dielectric **1851** may overlap with the first antenna element **1821** and the fifth antenna element **1831** when viewed from above the first surface **1811** of the printed circuit board **1810** (e.g., when viewed in the +z axis direction).

[0308] According to an embodiment, the second dielectric **1852** may be positioned corresponding to the second antenna element **1822** of the first antenna array **1820** and the sixth antenna element **1832** of the second antenna array **1830**. The second dielectric **1852** may overlap with the second antenna element **1822** and the sixth antenna element **1832** when viewed from above the first surface **1811** of the printed circuit board **1810** (e.g., when viewed in the +z axis direction).

[0309] According to an embodiment, the third dielectric **1853** may be positioned corresponding to the third antenna element **1823** of the first antenna array **1820** and the seventh antenna element **1833** of the second antenna array **1830**. The third dielectric **1853** may overlap with the third antenna element **1823** and the seventh antenna element **1833** when viewed from above the first surface **1811** of the printed circuit board **1810** (e.g., when viewed in the +z axis direction).

[0310] According to an embodiment, the fourth dielectric **1854** may be positioned corresponding to the fourth antenna element **1824** of the first antenna array **1820** and the eighth antenna element **1834** of the second antenna array **1830**. The fourth dielectric **1854** may overlap with the fourth antenna element **1824** and the eighth antenna element **1834** when viewed from above the first surface **1811** of the printed circuit board **1810** (e.g., when viewed in the +z axis direction).

[0311] According to an embodiment, the antenna module **18** may include a first electrical path (or a first transmission line or a first feeding line) (e.g., a first conductive path) (not shown) that transmits an electromagnetic signal (or a radio signal, an RF signal, or a radiated current) to the first dielectric **1851**. The first end of the first electrical path may be electrically connected to a communication circuit (e.g., the communication circuit **54** of FIG. **6**). For example, the second end of the first electrical path may be in physical contact with or directly connected to the first dielectric **1851**. The portion of the first dielectric **1851** to which the electromagnetic signal is transmitted from the second end of the first electrical path may be defined or interpreted as the first feeding portion **FP11**. In some embodiments, the first electrical path, or the second end of the first electrical path, may be defined or interpreted as a 'first feeding portion' and the portion of the first dielectric **1851** to which the electromagnetic signal is transmitted from the second end of the first electrical path may be defined or interpreted as a 'first feeding area' or a 'first feeding point'. The electromagnetic signal provided by the communication circuit (e.g., the communication circuit **54** of FIG. **6**) to the first feeding portion **FP11** (or the first electrical path, the first feeding area, or the

first feeding point) may be defined or interpreted as a ‘first feeding’. The remaining dielectrics **1852**, **1853**, and **1854** may include a first feeding portion (or a first feeding point) corresponding to the first feeding portion **FP11** of the first dielectric **1851**.

[0312] According to an embodiment, the first dielectric **1851** may operate as a first dielectric resonator (or first dielectric resonant antenna (DRA)) when the communication circuit (e.g., the communication circuit **54** of FIG. **6**) transmits (or feedings) an electromagnetic signal (or a radio signal, RF signal, or radiated current) through the first electrical path. When the communication circuit **54** transmits (or feedings) an electromagnetic signal through the first electrical path, a first dielectric resonator may be provided because of the discontinuity between the first dielectric **1851** and a medium in contact with the first dielectric **1851** (or around the first dielectric **1851**). The medium in contact with the first dielectric **1851** (or around the first dielectric **1851**), for example, may include air. When the first dielectric **1851** is in contact with the first cover **312** (see FIG. **3**), the medium in contact with the first dielectric **1851** (or around the first dielectric **1851**) may include the first cover **312**.

[0313] According to an embodiment, the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and the fourth dielectric **1854** may include a non-conductive material. With respect to a frequency (e.g., a resonant frequency) of the electromagnetic signal transmitted and/or received through the antenna module **18**, the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and the fourth dielectric **1854** may have a relative permittivity that is greater than that of a conductor having a relative permittivity being substantially 1.

[0314] According to an embodiment, the combination of the first dielectric resonator including the first dielectric **1851**, the resonator including the first antenna element **1821** (hereinafter referred to as a ‘first patch resonator’), and the resonator including the fifth antenna element **1831** (hereinafter referred to as a ‘fifth patch resonator’) may be defined or interpreted as the first radiating portion. In substantially the same manner as the first radiating portion, the antenna module **18** may include a second radiating portion provided by a combination of a second dielectric resonator including a second dielectric **1852**, a resonator including a second antenna element **1822** (hereinafter referred to as a ‘second patch resonator’), and a resonator including a sixth antenna element **1832** (hereinafter referred to as a ‘sixth patch resonator’). In substantially the same manner as the first radiating portion, the antenna module **18** may include a third radiating portion provided by a combination of a third dielectric resonator including a third dielectric **1853**, a resonator including a third antenna element **1823** (hereinafter referred to as a ‘third patch resonator’), and a resonator including a seventh antenna element **1833** (hereinafter referred to as a ‘seventh patch resonator’). In substantially the same manner as the first radiating portion, the antenna module **18** may include a fourth radiating portion provided by a combination of a fourth dielectric resonator including a fourth dielectric **1854**, a resonator including a fourth antenna element **1824** (hereinafter referred to as a ‘fourth patch resonator’), and a resonator including an eighth antenna element **1834** (hereinafter referred to as an ‘eight patch resonator’).

[0315] According to an embodiment, during the second feeding to the first radiating portion, a beam pattern of first linear polarization having a first polarization direction (e.g., the sixth direction **1801**) (or oscillating in the first polarization direction) may be radiated from the first patch resonator including the first antenna element **1821** of the first radiating portion. During the second feeding to the second radiating portion, a beam pattern of first linear polarization having the first polarization direction may be radiated from the second patch resonator including the second antenna element **1822** of the second radiating portion. During the second feeding to the third radiating portion, a beam pattern of first linear polarization having the first polarization direction may be radiated from the third patch resonator including the third antenna element **1823** of the third radiating portion. During the second feeding to the fourth radiating portion, a beam pattern of first linear polarization having the first polarization direction may be radiated from the fourth patch resonator including the fourth antenna element **1824** of the fourth radiating portion. During the

second feeding, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion, may provide a composite beam pattern (or a radiation pattern) in a first polarization direction that can radiate or concentrate relatively more wave energy for a coverage area corresponding to the first direction {circle around (1)} in which the first surface **1811** of the printed circuit board **1810** is facing.

[0316] According to an embodiment, during the third feeding to the first radiating portion, a beam pattern of second linear polarization having a second polarization direction (e.g., the seventh direction **1802**) (or oscillating in the second polarization direction) may be radiated from the first patch resonator including the first antenna element **1821** of the first radiating portion. During the third feeding to the second radiating portion, a beam pattern of second linear polarization having the second polarization direction may be radiated from the second patch resonator including the second antenna element **1822** of the second radiating portion. During the third feeding to the third radiating portion, a beam pattern of the second linear polarization having the second polarization direction may be radiated from the third patch resonator including the third antenna element **1823** of the third radiating portion. During the third feeding to the fourth radiating portion, a beam pattern of the second linear polarization having the second polarization direction may be radiated from the fourth patch resonator including the fourth antenna element **1824** of the fourth radiating portion. During the third feeding, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion may provide a composite beam pattern (or a radiation pattern) in a second polarization direction that can radiate or concentrate relatively more wave energy for a coverage area corresponding to the first direction {circle around (1)} in which the first surface **1811** of the printed circuit board **1810** is facing.

[0317] According to an embodiment, during the fourth feeding to the first radiating portion, a beam pattern of third linear polarization having a third polarization direction (or the second polarization direction) (e.g., the seventh direction **1802**) (or oscillating in the third polarization direction) may be radiated from the fifth patch resonator including the fifth antenna element **1831** of the first radiating portion. During the fourth feeding to the second radiating portion, a beam pattern of third linear polarization having the third polarization direction may be radiated from the sixth patch resonator including the sixth antenna element **1832** of the second radiating portion. During a fourth feeding to the third radiating portion, a beam pattern of a third linear polarization having a third polarization direction may be radiated from a seventh patch resonator including a seventh antenna element **1833** of the third radiating portion. During the fourth feeding to the fourth radiating portion, a beam pattern of the third linear polarization having the third polarization direction may be radiated from the eighth patch resonator including the eighth antenna element **1834** of the fourth radiating portion. During the fourth feeding, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion may provide a composite beam pattern (or radiation pattern) in the third polarization direction that can radiate or concentrate relatively more wave energy for a coverage area corresponding to the first direction {circle around (1)} in which the first surface **1811** of the printed circuit board **1810** is facing.

[0318] According to an embodiment, during a fifth feeding to the first radiating portion, a beam pattern of fourth linear polarization having a fourth polarization direction (or the first polarization direction) (e.g., the sixth direction **1801**) (or oscillating in the fourth polarization direction) may be radiated from the fifth patch resonator including the fifth antenna element **1831** of the first radiating portion. During a fifth feeding to the second radiating portion, a beam pattern of fourth

linear polarization having the fourth polarization direction may be radiated from a sixth patch resonator including a sixth antenna element **1832** of the second radiating portion. In a fifth feeding to the third radiating portion, a beam pattern of a fourth linear polarization having a fourth polarization direction may be radiated from a seventh patch resonator including a seventh antenna element **1833** of the third radiating portion. During a fifth feeding to the fourth radiating portion, a beam pattern of a fourth linear polarization having a fourth polarization direction may be radiated from an eighth patch resonator including the eighth antenna element **1834** of the fourth radiating portion. During the fifth feeding, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion may provide a composite beam pattern (or radiation pattern) in the fourth polarization direction that can radiate or concentrate relatively more wave energy for a coverage area corresponding to the first direction {circle around (1)} in which the first surface **1811** of the printed circuit board **1810** is facing.

[0319] According to an embodiment, during the first feeding and the second feeding to the first radiating portion, because of the electromagnetic coupling between the first dielectric resonator including the first dielectric **1851** and the first antenna element **1821**, a beam pattern of first linear polarization may be provided such that a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to a second direction {circle around (2)} or a third direction {circle around (3)} tilted with respect to the first direction {circle around (1)}. The second direction {circle around (2)} and third direction {circle around (3)} may be substantially perpendicular to the seventh direction **1802** (e.g., the y-axis direction), and may form an acute or obtuse angle with the first direction {circle around (1)} (e.g., the -z-axis direction). The second direction {circle around (2)} and third direction {circle around (3)} may be provided substantially symmetrically when viewed in the seventh direction **1802**. In an embodiment, a beam pattern of first linear polarization may be provided such that, during the first feeding and the second feeding to the first radiating portion with a -90° phase difference or a $+90^\circ$ phase difference, a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the second direction {circle around (2)} or a coverage area corresponding to the third direction {circle around (3)}. In substantially the same manner as the first radiating portion, a beam pattern of the first linear polarization may be provided in which a relatively more wave energy may be radiated or concentrated in a coverage area corresponding to the second direction {circle around (2)} or the third direction {circle around (3)} during the first feeding and the second feeding for the second radiating portion, the third radiating portion, and the fourth radiating portion. During the first feeding and the second feeding with a -90° phase difference or a $+90^\circ$ phase difference, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion may provide a composite beam pattern (or a radiation pattern) in the first polarization direction that can radiate or concentrate relatively more wave energy for a coverage area corresponding to the second direction {circle around (2)} or the third direction {circle around (3)}.

[0320] According to an embodiment, during the first feeding and the third feeding to the first radiating portion, because of electromagnetic coupling between the first dielectric resonator including the first dielectric **1851** of the first radiating portion and the first antenna element **1821**, a beam pattern of a second linear polarization may be provided such that a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth or the fifth direction tilted with respect to the first direction {circle around (1)}. The fourth direction and the fifth direction may be substantially parallel to the seventh direction **1802** (e.g., the y-axis direction), and may form an acute or obtuse angle with the first direction {circle around (1)} (e.g., the -z-axis direction). The fourth direction and the fifth direction may be provided substantially symmetrically

when viewed in the sixth direction **1801**. In an embodiment, a beam pattern of a second linear polarization may be provided such that, during the first feeding and the third feeding to the first radiating portion with a -90° phase difference or a $+90^\circ$ phase difference, a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth direction or a coverage area corresponding to the fifth direction. In substantially the same manner as the first radiating portion, a beam pattern of the second linear polarization may be provided in which a relatively more wave energy may be radiated or concentrated in the first feeding and the third feeding to the second radiating portion, the third radiating portion, and the fourth radiating portion for a coverage area corresponding to the fourth direction or the fifth direction. Interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiation portion, the beam pattern from the second radiation portion, the beam pattern from the third radiation portion, and the beam pattern from the fourth radiation portion during the first feeding and the third feeding with a -90° phase difference or a $+90^\circ$ phase difference may provide a composite beam pattern (or a radiation pattern) of the second polarization direction in which a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth direction or the fifth direction.

[0321] According to an embodiment, during the first feeding and the fourth feeding, because of electromagnetic coupling between the first dielectric resonator and the fifth antenna element **1831** including the first dielectric **1851** of the first radiating portion, a beam pattern of a third linear polarization may be provided such that a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth or the fifth direction tilted corresponding to the first direction. In an embodiment, a beam pattern of the third linear polarization may be provided such that, during the first feeding and the fourth feeding of the first radiating portion with a -90° phase difference or a $+90^\circ$ phase difference, a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth direction or a coverage area corresponding to the fifth direction. In substantially the same manner as the first radiating portion, a beam pattern of the third linear polarization may be provided in which a relatively more wave energy may be radiated or concentrated in the first feeding and the fourth feeding to the second radiating portion, the third radiating portion, and the fourth radiating portion in a coverage range corresponding to the fourth direction or the fifth direction. Interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiation portion, the beam pattern from the second radiation portion, the beam pattern from the third radiation portion, and the beam pattern from the fourth radiation portion during the first feeding and the fourth feeding with a -90° phase difference or a $+90^\circ$ phase difference may provide a composite beam pattern (or a radiation pattern) of the third polarization direction from which a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to the fourth direction or the fifth direction.

[0322] According to an embodiment, during the first feeding and the fifth feeding to the first radiating portion, because of electromagnetic coupling between the first dielectric resonator including the first dielectric **1851** of the first radiating portion and the fifth antenna element **1831**, a beam pattern of a fourth linear polarization may be provided such that a relatively more wave energy may be radiated or concentrated for a coverage area corresponding to a second direction {circle around (2)} or a third direction {circle around (3)} tilted corresponding to the first direction {circle around (1)}. In an embodiment, a beam pattern of the fourth linear polarization may be provided in which the relatively more wave energy may be radiated or concentrated for the coverage area corresponding to the second direction {circle around (2)} or the coverage area corresponding to the third direction {circle around (3)} during the first feeding and the fifth feeding to the first radiating portion with a -90° phase difference or a $+90^\circ$ phase difference. In substantially the same manner as the first radiating portion, a beam pattern of the fourth linear polarization may be provided in which a relatively more wave energy may be radiated or

concentrated for a coverage area corresponding to the second direction {circle around (2)} or the third direction {circle around (3)} during the first feeding and the fifth feeding for the second radiating portion, the third radiating portion, and the fourth radiating portion. During the first feeding and the fifth feeding with a -90° phase difference or a $+90^\circ$ phase difference, interference (e.g., constructive interference and/or destructive interference) between the beam pattern from the first radiating portion, the beam pattern from the second radiating portion, the beam pattern from the third radiating portion, and the beam pattern from the fourth radiating portion may provide a composite beam pattern (or a radiation pattern) in the fourth polarization direction that may radiate or concentrate relatively more wave energy for a coverage area corresponding to the second direction {circle around (2)} or the third direction {circle around (3)}.

[0323] According to an embodiment, electromagnetic waves (e.g., linear polarization) radiated from the first radiating portion, the second radiating portion, the third radiating portion, or the fourth radiating portion during the second feeding, the first feeding and the second feeding, or the first feeding and the third feeding, may have a first resonant frequency included in the first frequency band. The electromagnetic waves (e.g., linear polarization) radiating from the first radiating portion, the second radiating portion, the third radiating portion, and the fourth radiating portion during the fourth feeding, the first feeding and the fourth feeding, the fifth feeding, or the first feeding and the fifth feeding, may have a second resonant frequency included in a second frequency band that is different from the first frequency band.

[0324] According to an embodiment, the printed circuit board **1810** may include a first conductive via **V12** disposed corresponding to the first radiating portion. The first conductive via **V12** may penetrate an opening **1821a** provided in a first antenna element **1821** included in the first radiating portion and an opening **1831a** provided in a fifth antenna element **1831** included in the first radiating portion. The first conductive via **V12** may be part of a first electrical path (or a first conductive path) connecting the first dielectric **1851** and a communication circuit (e.g., the communication circuit **54** of FIG. 6).

[0325] According to an embodiment, the antenna module **18** may include a first connector **1861** (e.g., the first connector **1231** of FIG. 12) disposed at least partially between the first dielectric **1851** and a first surface **1811** of the printed circuit board **1810** in response to the first dielectric **1851**. The first connector **1861** may include a first conductive portion **C12** (e.g., the first conductive portion **C12** of FIG. 12) connecting the first conductive via **V12** and the first dielectric **1851**. One end of the first conductive portion **C12** may be electrically connected to the first conductive via **V12** through a conductive adhesive material or a conductive bonding material such as solder. The first conductive portion **C12** may be part of a first electrical path (or a first conductive path) connecting the first dielectric **1851** and a communication circuit (e.g., the communication circuit **54** of FIG. 6).

[0326] According to an embodiment, the other end of the first conductive portion **C12** may be in physical contact or direct connection with the first dielectric **1851**. During the first feeding by a communication circuit (e.g., the communication circuit **54** of FIG. 6), the first dielectric **1851** may receive electromagnetic signals directly from the first conductive portion **C12**.

[0327] According to an embodiment, the first connector **1861** may include a first non-conductive portion **18611** (e.g., the first non-conductive portion **12311** of FIG. 14) including a non-conductive material such as a polymer. The first conductive portion **C12** may be disposed on the first non-conductive portion **18611**.

[0328] According to an embodiment, the first connector **1861** may be implemented as a printed circuit board. For example, the first non-conductive portion **18611** may be a non-conductive layer (e.g., a dielectric layer) of a printed circuit board, and the first conductive portion **C12** may be a conductive via provided on the non-conductive layer.

[0329] According to various embodiments, the first conductive portion **C12** may be connected to the first dielectric **1851** indirectly through the first non-conductive portion **18611**. The first

dielectric **1851** and the first non-conductive portion **18611** may be in physical contact. During the first feeding by a communication circuit (e.g., the communication circuit **54** of FIG. **6**), the first dielectric **1851** may indirectly receive electromagnetic signals from the first conductive portion **C12** through the first non-conductive portion **18611**. In various embodiments, the first dielectric resonator may be interpreted to include a combination of the first dielectric **1851** and the first non-conducting portion **18611**.

[0330] According to various embodiments, the antenna module **18** may include a non-conductive adhesive material or a non-conductive bonding material disposed between the first dielectric **1851** and the first non-conductive portion **18611**. The first dielectric **1851** and the first non-conductive portion **18611** may be coupled through the non-conductive adhesive material or the non-conductive bonding material. During the first feeding by a communication circuit (e.g., the communication circuit **54** of FIG. **6**), the first dielectric **1851** may indirectly receive electromagnetic signals from the first conductive portion **C12** through the first non-conductive portion **18611** and the non-conductive adhesive material (or the non-conductive bonding material). In various embodiments, the first dielectric resonator may be interpreted to include a combination of the first dielectric **1851**, the first non-conductive portion **18611**, and the non-conductive adhesive material (or the non-conductive bonding material).

[0331] According to an embodiment, the antenna module **18** may include a second connector (e.g., the second connector **1232** of FIG. **12**) disposed at least partially between the second dielectric **1852** and the first surface **1811** of the printed circuit board **1810** in response to the second dielectric **1852**. The antenna module **18** may include a third connector (e.g., the third connector **1233** of FIG. **12**) disposed at least partially between the third dielectric **1853** and the first surface **1811** of the printed circuit board **1810** in response to the third dielectric **1853**. The antenna module **18** may include a fourth connector (e.g., the fourth connector **1234** of FIG. **12**) disposed at least partially between the fourth dielectric **1854** and the first surface **1811** of the printed circuit board **1810** in response to the fourth dielectric **1854**. The second connector, third connector, or fourth connector may be provided substantially the same as or similar to the first connector **1861** connecting the printed circuit board **1810** and the first dielectric **1851**.

[0332] According to various embodiments, in a variation of the embodiment of FIG. **19**, the antenna module **18** may be implemented with the first dielectric **1861**, the second dielectric, the third dielectric, and the fourth dielectric omitted, as in the embodiment of FIG. **15**.

[0333] According to various embodiments, the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and the fourth dielectric **1854** may be a plurality of non-conductive material members (not shown separately) disposed on the first cover **312** (see FIG. **3**) of the foldable electronic device **3**.

[0334] According to various embodiments, the first cover **312** (see FIG. **3**) of the foldable electronic device **3** may replace the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and the fourth dielectric **1854**.

[0335] FIG. **20** are heat maps illustrating wave energy for electromagnetic waves radiated by various feedings to the antenna module **18** of FIG. **18** according to an embodiment of the disclosure.

[0336] Referring to FIG. **20**, during a feeding to at least one radiating portion provided by a combination of dielectric resonators and patch resonators included in the antenna module **18**, a beam pattern that can achieve transmission and reception performance for electromagnetic signals in coverage areas that would otherwise be difficult to cover by a beam pattern radiated by a feeding to at least one patch resonator included in the antenna module **18**, may be provided.

[0337] FIG. **21** is a block diagram illustrating a communication system **21** included in an electronic device (e.g., the foldable electronic device **3** of FIG. **3**) according to an embodiment of the disclosure.

[0338] Referring to FIG. **21**, the communication system **21** may include at least one dielectric

2110, a first RF chain **2121**, a second RF chain **2122**, a third RF chain **2123**, a fourth RF chain **2124**, a first matching circuit **2131**, a second matching circuit **2132**, a third matching circuit **2133**, and/or a fourth matching circuit **2134**.

[0339] According to an embodiment, the at least one dielectric **2110** may be implemented to act as a dielectric resonator when fed through an electrical path (hereinafter, a ‘dielectric feeding path’) **2111** connected to the at least one dielectric **2110**. The dielectric **2110** may include, for example, the first dielectric **1851**, the second dielectric **1852**, the third dielectric **1853**, and/or the fourth dielectric **1854** of FIG. **18**.

[0340] According to an embodiment, the first RF chain **2121** may be implemented to transmit and/or receive electromagnetic signals of a first linear polarization having a first polarization direction through at least one antenna radiator included in an electronic device (e.g., the foldable electronic device **3** of FIG. **3**) in a first frequency band of use (or a first operating frequency band). In response to the first RF chain **2121**, the position of the feeding portion (or the feeding point) with respect to the at least one antenna radiator (or on the at least one antenna radiator) may be provided such that the first linear polarization having the first polarization direction may be radiated through the at least one antenna radiator.

[0341] According to an embodiment, the second RF chain **2122** may be implemented to transmit and/or receive electromagnetic signals of a second linear polarization having a second polarization direction perpendicular to the first polarization direction through the at least one antenna radiator included in the electronic device (e.g., the foldable electronic device **3** of FIG. **3**) in a first frequency band of use (or a first operating frequency band). In response to the second RF chain **2122**, the position of the feeding portion (or the feeding point) with respect to the at least one antenna radiator (or on the at least one antenna radiator) may be provided such that the second linear polarization having the second polarization direction may be radiated through the at least one antenna radiator.

[0342] According to an embodiment, the third RF chain **2123** may be implemented to transmit and/or receive electromagnetic signals of the third linear polarization having the first polarization direction through the at least one antenna radiator included in the electronic device (e.g., the foldable electronic device **3** of FIG. **3**) in a second frequency band of use (or a second operating frequency band) that is different from the first frequency band. Corresponding to the third RF chain **2123**, the position of the feeding portion (or the feeding point) to the at least one antenna radiator (or on the at least one antenna radiator) may be provided such that the third linear polarization having the first polarization direction may be radiated through the at least one antenna radiator.

[0343] According to an embodiment, the fourth RF chain **2124** may be implemented to transmit and/or receive electromagnetic signals of the fourth linear polarization having the second polarization direction through the at least one antenna radiator included in the electronic device (e.g., the foldable electronic device **3** of FIG. **3**) in a second frequency band of use (or a second operating frequency band). Corresponding to the fourth RF chain **2124**, the position of the feeding portion (or the feeding point) to the at least one antenna radiator (or on the at least one antenna radiator) may be provided such that the fourth linear polarization having the second polarization direction may be radiated through the at least one antenna radiator.

[0344] According to an embodiment, the first RF chain **2121**, the second RF chain **2122**, the third RF chain **2123**, and the fourth RF chain **2124** may be disposed on another printed circuit board electrically connected to the antenna module **18** of FIG. **18** (e.g., a printed circuit board on which components such as the processor **120**, the memory **130**, or the communication module **190** of FIG. **1** are disposed). A communication circuit (e.g., the communication circuit **54** of FIG. **6**) (hereinafter, a ‘second communication circuit’) included in antenna module **18** of FIG. **18** may exchange signals with another communication circuit (e.g., the wireless communication module **192** of FIG. **1**) (hereinafter, a ‘first communication circuit’) disposed on another printed circuit board. In an embodiment, the first RF chain **2121**, the second RF chain **2122**, the third RF chain

2123, and the fourth RF chain **2124** may be at least partially included in the first communication circuit.

[0345] According to an embodiment, the communication system **21** may include a switching circuit. The switching circuit may be included in the first communication circuit or the second communication circuit. In various embodiments, the switching circuit may be provided separately from the first communication circuit and the second communication circuit. The switching circuit may selectively electrically connect the first RF chain **2121**, the second RF chain **2122**, the third RF chain **2123**, or the fourth RF chain **2124** to the dielectric feeding path **2111** under the control of a processor (e.g., the processor **120** of FIG. 1).

[0346] According to an embodiment, when the first RF chain **2121** is electrically connected to the dielectric feeding path **2111** by a switching circuit, the at least one dielectric **2110** may act as a dielectric resonator to electromagnetically influence the first linear polarization to reduce degradation of transmission and/or reception performance.

[0347] According to an embodiment, when the second RF chain **2122** is electrically connected to the dielectric feeding path **2111** by a switching circuit, the at least one dielectric **2110** may act as a dielectric resonator to electromagnetically influence the second linear polarization to reduce degradation of transmission and/or reception performance for the second linear polarization.

[0348] According to an embodiment, when the third RF chain **2123** is electrically connected to the dielectric feeding path **2111** by a switching circuit, the at least one dielectric **2110** may act as a dielectric resonator to electromagnetically influence the third RF chain **2123** to reduce degradation of transmission and/or reception performance for the third linear polarization.

[0349] According to an embodiment, when the fourth RF chain **2124** is electrically connected to the dielectric feeding path **2111** by a switching circuit, the at least one dielectric **2110** may act as a dielectric resonator to electromagnetically influence the fourth RF chain **2124** to reduce degradation of transmission and/or reception performance for the fourth linear polarization.

[0350] According to an embodiment, when the first RF chain **2121** is configured to be electrically connected to the dielectric feeding path **2111**, the first RF chain **2121** may be electrically connected to the dielectric feeding path **2111** through the first matching circuit **2131**. When the second RF chain **2122** is configured to be electrically connected to the dielectric feeding path **2111**, the second RF chain **2122** may be electrically connected to the dielectric feeding path **2111** through the second matching circuit **2132**. When the third RF chain **2123** is configured to be electrically connected to the dielectric feeding path **2111**, the third RF chain **2123** may be electrically connected to the dielectric feeding path **2111** through the third matching circuit **2133**. When the fourth RF chain **2124** is configured to be electrically connected to the dielectric feeding path **2111**, the fourth RF chain **2124** may be electrically connected to the dielectric feeding path **2111** through the fourth matching circuit **2134**.

[0351] According to an embodiment, the matching circuit (e.g., the first matching circuit **2131**, the second matching circuit **2132**, the third matching circuit **2133**, or the fourth matching circuit **2134**) may include at least one electrical element having a component such as a capacitance, inductance, or conductance (e.g., a lumped element or a passive element), or a circuit implemented with a combination of these electrical elements.

[0352] According to an embodiment, the matching circuit (e.g., the first matching circuit **2131**, the second matching circuit **2132**, the third matching circuit **2133**, or the fourth matching circuit **2134**) may be tunable to a resonant frequency. The matching circuit may, for example, shift the resonant frequency to a specified frequency or shift the resonant frequency by a specified amount.

[0353] According to an embodiment, the matching circuit (e.g., the first matching circuit **2131**, the second matching circuit **2132**, the third matching circuit **2133**, or the fourth matching circuit **2134**) may match impedances. The matching circuit may reduce power dissipation at the selected or specified frequency to enable efficient signal transmission. The matching circuit may reduce the amount of reflections at the selected or specified frequency.

[0354] According to an embodiment, the first matching circuit **2131** and/or the second matching circuit **2132** may contribute to isolation to reduce electromagnetic effects between the first RF chain **2121** transmitting and/or receiving the first linear polarization and the second RF chain **2122** transmitting and/or receiving the second linear polarization.

[0355] According to an embodiment, the third matching circuit **2133** and/or the fourth matching circuit **2134** may contribute to isolation to reduce electromagnetic effects between the third RF chain **2123** transmitting and/or receiving the third linear polarization and the fourth RF chain **2124** transmitting and/or receiving the fourth linear polarization.

[0356] According to an embodiment, the first matching circuit **2131**, the second matching circuit **2132**, the third matching circuit **2133**, and/or the fourth matching circuit **2134** may be disposed on another printed circuit board electrically connected to the antenna module **18** of FIG. **18**, such as a printed circuit board on which components such as the processor **120**, the memory **130**, or the communication module **190** of FIG. **1** are disposed. In various embodiments, the first matching circuit **2131**, the second matching circuit **2132**, the third matching circuit **2133**, and/or the fourth matching circuit **2134** may be included in a second communication circuit disposed on another printed circuit board.

[0357] According to various embodiments, one matching circuit **2130A** may be implemented to replace the first matching circuit **2131** and the second matching circuit **2132** in response to the first frequency band of use. Another matching circuit **2130B** that replaces the third matching circuit **2133** and the fourth matching circuit **2134** in response to the second frequency band of use, may be implemented. A switching circuit included in the communication system **21** may, under the control of a processor (e.g., the processor **120** of FIG. **1**), electrically connect the two matching circuits **2130A** and **2130B** selectively to the dielectric feeding path **2111**.

[0358] FIG. **22** is a cross-sectional view illustrating a portion of an electronic device (e.g., the foldable electronic device **3** of FIG. **3**) according to an embodiment of the disclosure.

[0359] Referring to FIG. **22**, the electronic device **22A** according to a first embodiment may include an antenna structure **2211** and a dielectric **2212**. The antenna structure **2211** may be, for example, the antenna structure **51** of FIG. **5** or the antenna structure **181** of FIG. **18**. The dielectric **2212** may be positioned facing antenna structure **2211**. In an embodiment, the dielectric **2212** may be coupled to the antenna structure **2211**. The dielectric **2212**, for example, may be positioned in a direction in which the antenna structure **2211** substantially radiates electromagnetic waves during the feeding.

[0360] According to an embodiment, the dielectric **2212** may be electromagnetically coupled to the antenna structure **2211** during the feeding to operate as a dielectric resonator to extend coverage. The dielectric **2212** may receive electromagnetic signals from a communication circuit disposed on or electrically connected to antenna structure **2211**, such as communication circuit **54** of FIG. **6**.

[0361] According to an embodiment, the dielectric **2212** may include a non-conductive material (e.g., a polymer). The dielectric **2212** may have a relative permittivity that is greater than a conductor having a relative permittivity being substantially 1.

[0362] According to various embodiments, the dielectric **2212** may be included in a housing that provides an exterior surface of the electronic device **22A**. The dielectric **2212** may be, for example, the first cover **312** of the foldable electronic device **3** of FIG. **3**.

[0363] According to an embodiment, in the electronic device **22A** according to the first embodiment, the portion of the dielectric **2212** that faces the antenna structure **2211** or is in physical contact with (or coupled to) the antenna structure **2211** may have a relatively greater thickness. When viewed from above the dielectric **2212**, the portion of the dielectric **2212** that overlaps with the antenna element included in the antenna structure **2211** may have a relatively greater thickness.

[0364] The electronic device **22B** according to another second embodiment may include the antenna structure **2221**, the dielectric **2222**, and/or the metal body **2223**.

[0365] The antenna structure **2221** may be, for example, the antenna structure **51** of FIG. **5** or the antenna structure **181** of FIG. **18**. The dielectric **2222** may be coupled to the antenna structure **2221** or may be positioned facing the antenna structure **2221**. In an embodiment, the dielectric **2222** may be coupled to the antenna structure **2221**. The dielectric **2222**, for example, may be positioned in a direction in which the antenna structure **2221** substantially radiates electromagnetic waves during the feeding.

[0366] According to an embodiment, the dielectric **2222** may be electromagnetically coupled to the antenna structure **2221** during the feeding to operate as a dielectric resonator to extend coverage. The dielectric **2222** may receive electromagnetic signals from a communication circuit disposed on or electrically connected to the antenna structure **2221** (e.g., the communication circuit **54** of FIG. **6**).

[0367] According to an embodiment, the dielectric **2222** may include a non-conductive material (e.g., a polymer). The dielectric **2222** may have a relative permittivity that is greater than a conductor having a relative permittivity being substantially 1.

[0368] According to various embodiments, the dielectric **2222** may be included in a housing that provides an exterior surface of the electronic device **22B**. The dielectric **2222** may be, for example, the first cover **312** of the foldable electronic device **3** of FIG. **3**.

[0369] According to an embodiment, the metal body **2223** may be located on the interior of the electronic device **22B**. The metal body **2223** may be disposed to at least partially surround the antenna structure **2221**, as viewed from above the dielectric **2222**. The metal body **2223** may include, for example, openings or recesses, and the antenna structure **2221** may be inserted into the openings or recesses of the metal body **2223**. In various embodiments, the metal body **2223** may be coupled to the antenna structure **2221** and/or the dielectric **2222**.

[0370] The electronic device **22C** according to yet another third embodiment may include an antenna structure **2231**, a dielectric **2232**, and/or a metal body **2233**.

[0371] The antenna structure **2231** may be, for example, the antenna structure **51** of FIG. **5** or the antenna structure **181** of FIG. **18**. The dielectric **2232** may be positioned face to face with antenna structure **2231**. The dielectric **2232** may be coupled to the antenna structure **2231**. The dielectric **2232**, for example, may be positioned in a direction in which the antenna structure **2231** substantially radiates electromagnetic waves during the feeding.

[0372] According to an embodiment, the dielectric **2232** may be electromagnetically coupled to the antenna structure **2231** during the feeding to operate as a dielectric resonator to extend coverage. The dielectric **2232** may receive electromagnetic signals from a communication circuit disposed on or electrically connected to the antenna structure **2231**, such as the communication circuit **54** of FIG. **6**.

[0373] According to an embodiment, the dielectric **2232** may include a non-conductive material (e.g., a polymer). The dielectric **2232** may have a relative permittivity that is greater than a conductor having a relative permittivity being substantially 1.

[0374] According to an embodiment, the metal body **2233** may be disposed to at least partially surround the antenna structure **2231** and/or the dielectric **2232**, as viewed from above the dielectric **2232**. The metal body **2233** may include, for example, openings or recesses, and the antenna structure **2231** and dielectric **2232** may be inserted into the openings or recesses of the metal body **2233**. In various embodiments, the metal body **2233** may be coupled to the antenna structure **2231** and/or the dielectric **2232**.

[0375] According to various embodiments, the dielectric **2232** and the metal body **2233** may be included in a housing that provides an exterior surface of the electronic device **22C**. The dielectric **2232** may be, for example, a portion included in the first cover **312** of the foldable electronic device **3** of FIG. **3**, and the metal body **2233** may be the other portion included in the first cover **312**.

[0376] According to an aspect of the disclosure, an electronic device (e.g., the foldable electronic

device **3**) is provided. The electronic device includes a wireless communication circuit (e.g., a communication circuit **54**), an antenna structure **51** including a printed circuit board **510** electrically connected to the wireless communication circuit (e.g., the communication circuit **54**), wherein the printed circuit board **510** includes a first surface **511**, a second surface **512** facing an opposite direction from the first surface **511**, and an antenna element (e.g., a first antenna element **521**) disposed on the first surface **511** or disposed inside the printed circuit board **510** closer to the first surface **511** than the second surface **512**, a non-conductive dielectric (e.g., a first dielectric **61**) facing the first surface **511** and overlapping with the antenna element (e.g., the first antenna element **521**) when viewed from above the first surface **511**, and a conductive path at least a portion of which is disposed on the printed circuit board **510** and which is electrically connected to the wireless communication circuit (e.g., the communication circuit **54**), wherein the conductive path is in physical contact with a non-conductive dielectric (e.g., a first dielectric **61**).

[0377] According to an example embodiment of the disclosure, the conductive path is physically separated from an antenna element (e.g., the first antenna element **521**).

[0378] According to an example embodiment of the disclosure, the conductive path penetrates an opening **5211** provided in the antenna element (e.g., the first antenna element **521**).

[0379] According to an example embodiment of the disclosure, the conductive path includes a conductive via (e.g., a first conductive via **V1**) included in the printed circuit board **510**.

[0380] According to an example embodiment of the disclosure, the electronic device **3** may further include a connector (e.g., a first connector **1231**) disposed between a non-conductive dielectric **1221** and the first surface **1210a**, wherein the conductive path includes a conductive via (e.g., a first conductive via **V1**) included in the printed circuit board **1210**, and a conductive portion (e.g., a first conductive portion **C1**) included in the connector (e.g., the first connector **1231**) so as to be electrically connected to the conductive via (e.g., the first conductive via **V1**), and wherein the conductive portion (e.g., the first conductive portion **C1**) is in physical contact with the non-conductive dielectric **1221**.

[0381] According to example embodiments of the disclosure, the connector (e.g., the first connectors **1231**) is another printed circuit board, and a conductive portion (e.g., the first conductive portion **C1**) is a conductive via provided on the other printed circuit board.

[0382] According to an example embodiment of the disclosure, the printed circuit board **1810** further includes another antenna element (e.g., a fifth antenna element **1831**) disposed closer to the first surface **1811** than the antenna elements (e.g., the first antenna element **1821**).

[0383] According to an example embodiment of the disclosure, a wireless communication circuit (e.g., a communication circuit **54**) is disposed on the second surface **512**.

[0384] According to an example embodiment of the disclosure, the wireless communication circuit (e.g., the communication circuit **54**) varies a phase of the electromagnetic signal provided to the antenna element (e.g., a first antenna element **521**) and a phase of the electromagnetic signal provided to the conductive path.

[0385] According to an example embodiment of the disclosure, the printed circuit board **510** further includes a ground plane **53** disposed closer to the second surface **512** than the antenna element (e.g., the first antenna element **521**).

[0386] According to an example embodiment of the disclosure, the electronic device (e.g., the foldable electronic device **3**) further includes another wireless communication circuit and a switching circuit disposed in an electrical path between the conductive path and the other wireless communication circuit.

[0387] According to an example embodiment of the disclosure, the electronic device (e.g., the foldable electronic device **3**) further includes a matching circuit (e.g., the first matching circuit **2131**) disposed in an electrical path between the conductive path and the other wireless communication circuit.

[0388] According to an example embodiment of the disclosure, the portion of the non-conductive

dielectric **2212** overlapping with the antenna element (e.g., the first antenna element **521**) has a relatively greater thickness, when viewed from above the first surface **511**.

[0389] According to an example embodiment of the disclosure, the electronic device (e.g., the foldable electronic device **3**) further includes a cover (e.g., the first cover **312**) providing an exterior surface of the electronic device (e.g., the foldable electronic device **3**), wherein the non-conductive dielectric (e.g., a first dielectric **61**) is positioned between the first surface **511** and the cover (e.g., the first cover **312**).

[0390] According to an example embodiment of the disclosure, wherein the non-conductive dielectric (e.g., the first dielectric **61**) is included in a housing (e.g., the foldable housing **30**) that provides an appearance of an electronic device (e.g., the foldable electronic device **3**).

[0391] According to an example embodiment of the disclosure, the electronic device **3** may include a wireless communication circuit **54**, an antenna structure **181**, a non-conductive dielectric (e.g., a first dielectric **1851**), and a conductive path. The antenna structure **181** may include a printed circuit board **1810** electrically connected to the wireless communication circuit **54**. The printed circuit board **1810** may include a first surface **1811** and a second surface **1812** facing in an opposite direction from the first surface **1811**. The printed circuit board **1810** may include a first antenna element **1821** disposed on the first surface **1811**, or disposed within the interior of the printed circuit board **1810** closer to the first surface **1811** than the second surface **1812**. The printed circuit board **1810** may include a second antenna element (e.g., a fifth antenna element **1831**) disposed closer to the first surface **1811** than the first antenna element **1821**. A non-conductive dielectric (e.g., a first dielectric **1851**) may face the first surface **1811** and may overlap with the first antenna element **1821** and the second antenna element (e.g., the fifth antenna element **1831**) when viewed from above the first surface **1811**. The conductive paths may be disposed at least partially on the printed circuit board **1810** and may be electrically connected to the wireless communication circuit **54**. The conductive path may be in physical contact with a non-conductive dielectric (e.g., a first dielectric **1851**).

[0392] According to an example embodiment of the disclosure, the conductive path may be physically separated from the first antenna element **1821** and the second antenna element (e.g., the fifth antenna element **1831**).

[0393] According to an example embodiment of the disclosure, the conductive path may penetrate an opening **1821a** provided in the first antenna element **1821** and an opening **1831a** provided in the second antenna element (e.g., the fifth antenna element **1831**).

[0394] According to an example embodiment of the disclosure, the conductive path may include a conductive via (e.g., a first conductive via **V12**) included in the printed circuit board **1810**.

[0395] According to an example embodiment of the disclosure, the electronic device (e.g., the foldable electronic device **3**) may further include a connection (e.g., a first connector **1861**) disposed between a non-conductive dielectric (e.g., a first dielectric **1851**) and the first surface **1811**. The conductive path may include a conductive via (e.g., first conductive via **V12**) included in the printed circuit board **1810**, and a conductive portion (e.g., a first conductive portion **C12**) included in the connector (e.g., the first connector **1861**) for electrical connection with the conductive via (e.g., the first conductive via **V12**). The conductive portion (e.g., the first conductive portion **C12**) may be in physical contact with a non-conductive dielectric (e.g., the first dielectric **1851**).

[0396] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure, as defined by the appended claims and their equivalents.

Claims

- 1.** An electronic device comprising: a wireless communication circuit; an antenna structure including a printed circuit board electrically connected to the wireless communication circuit, wherein the printed circuit board includes a first surface, a second surface facing an opposite direction from the first surface, and an antenna element disposed on the first surface or disposed inside the printed circuit board closer to the first surface than the second surface; a non-conductive dielectric facing the first surface and overlapping with the antenna element when viewed from above the first surface; and a conductive path at least a portion of which is disposed on the printed circuit board and which is electrically connected to the wireless communication circuit, wherein the conductive path is in physical contact with the non-conductive dielectric.
 - 2.** The electronic device of claim 1, wherein the conductive path is physically separated from the antenna element.
 - 3.** The electronic device of claim 1, wherein the conductive path penetrates an opening provided in the antenna element.
 - 4.** The electronic device of claim 1, wherein the conductive path comprises a conductive via included in the printed circuit board.
 - 5.** The electronic device of claim 1, further comprising a connector disposed between a conductive dielectric and the first surface, wherein the conductive dielectric comprises a conductive via included in the printed circuit board and a conductive portion included in the connector so as to be electrically connected to the conductive via, and wherein the conductive portion is in physical contact with the non-conductive dielectric.
 - 6.** The electronic device of claim 5, wherein the connector is another printed circuit board, and the conductive portion is a conductive via provided on the other printed circuit board.
 - 7.** The electronic device of claim 1, wherein the printed circuit board further comprises another antenna element disposed closer to the first surface than the antenna element, and wherein the antenna element and the other antenna element overlap each other when viewed from above the first surface.
 - 8.** The electronic device of claim 1, wherein the wireless communication circuit is disposed on the second surface.
 - 9.** The electronic device of claim 1, wherein the wireless communication circuit varies an electromagnetic signal provided to the antenna element and a phase of an electromagnetic signal provided to the conductive path.
 - 10.** The electronic device of claim 1, wherein the printed circuit board further comprises a ground plane disposed closer to the second surface than the antenna element.
 - 11.** The electronic device of claim 1, further comprising another wireless communication circuit and a switching circuit disposed in an electrical path between the conductive path and the other wireless communication circuit.
 - 12.** The electronic device of claim 11, further comprising a matching circuit disposed in an electrical path between the conductive path and the other wireless communication circuit.
 - 13.** The electronic device of claim 1, wherein, the portion of the non-conductive dielectric overlapping with the antenna element has a relatively greater thickness, when viewed from above the first surface.
 - 14.** The electronic device of claim 1, further comprising a cover providing an outer surface of the electronic device, wherein the non-conductive dielectric is positioned between the first surface and the cover.
 - 15.** The electronic device of claim 1, wherein, the non-conductive dielectric is included in a housing that provides an appearance of the electronic device.
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